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AIME Algebra Problem Collection

New mathematical restatements of the standard 1983–2026 archive. Contest credit: MAA American Mathematics Competitions. Linked sources supply official wording and diagrams.

1983 AIME Problem 1: Reciprocal logarithms

For x,y,z>1 and w>0, suppose log_x(w)=24, log_y(w)=40, and log_(xyz)(w)=12. Find log_z(w).

Published problem and solutions · MAA AMC.

1983 AIME Problem 2: Absolute values on a fixed interval

Let 0<p<15 and f(x)=|x-p|+|x-15|+|x-p-15|. Find the minimum of f on p≤x≤15.

Published problem and solutions · MAA AMC.

1983 AIME Problem 3: A radical equation with a repeated quadratic

Find the product of all real roots of $x^2+18x+30=2\sqrt{(x^2+18x+45)}$.

Published problem and solutions · MAA AMC.

1983 AIME Problem 5: Recovering a sum from power sums

Complex numbers x,y satisfy $x^2+y^2=7$ and $x^3+y^3=10$. Find the largest real value of x+y.

Published problem and solutions · MAA AMC.

1983 AIME Problem 9: An attainable inequality bound

Find the minimum of $(9x^2\sin^2 x+4)/(x \sin x)$ for $0<x<\pi$.

Published problem and solutions · MAA AMC.

1983 AIME Problem 13: Sum of alternating subset sums

For every nonempty subset of 1 through 7, list its elements decreasingly and alternately add and subtract them, starting with plus. Sum these values over all subsets.

Published problem and solutions · MAA AMC.

1984 AIME Problem 1: The even-indexed half of a progression

An arithmetic progression has common difference 1 and $a_1+\cdots+a_{98}=137$. Find $a_2+a_4+\cdots+a_{98}$.

Published problem and solutions · MAA AMC.

1984 AIME Problem 4: Recovering a list from two means

A list of positive integers contains 68 and has mean 56. Removing one occurrence of 68 leaves mean 55. Find the largest possible entry in the original list.

Published problem and solutions · MAA AMC.

1984 AIME Problem 5: A product from symmetric logarithms

Positive a,b satisfy $\log_8 a+\log_4(b^2)=5$ and $\log_8 b+\log_4(a^2)=7$. Find ab.

Published problem and solutions · MAA AMC.

1984 AIME Problem 7: A nested recurrence determined by parity

For every integer n, let f(n)=n-3 when $n\geq 1000$ and f(n)=f(f(n+5)) when $n<1000$. Find f(84).

Published problem and solutions · MAA AMC.

1984 AIME Problem 8: A restricted ninth root of unity

A root z of $z^6+z^3+1=0$ has argument θ strictly between 90° and 180°. Find θ in degrees.

Published problem and solutions · MAA AMC.

1984 AIME Problem 10: A score that determines the number correct

A 30-question test scores 30+4c−w, with c correct, w wrong and the rest blank. Find the smallest score above 80 that uniquely determines c.

Published problem and solutions · MAA AMC.

1984 AIME Problem 12: Root orbits under two reflections

A real function is unchanged by reflection about 2 and about 7, and vanishes at 0. Find the fewest roots it can have in [−1000,1000].

Published problem and solutions · MAA AMC.

1984 AIME Problem 13: A telescoping inverse-trigonometric sum

Evaluate 10 cot(arccot 3+arccot 7+arccot 13+arccot 21), taking each inverse cotangent in$(0,\pi/2)$.

Published problem and solutions · MAA AMC.

1984 AIME Problem 15: Extracting a coefficient from a rational system

Real x,y,z,w satisfy $x^2/(t-1)+y^2/(t-9)+z^2/(t-25)+w^2/(t-49)=1$ at t=4,16,36,64. Find $x^2+y^2+z^2+w^2$.

Published problem and solutions · MAA AMC.

1985 AIME Problem 1: Pairing a reciprocal recurrence

Let $x_1=97$ and $x_n=n/x_{n-1}$ for n>1. Find $x_1x_2\cdots x_8$.

Published problem and solutions · MAA AMC.

1985 AIME Problem 3: An integer complex cube

Positive integers a,b,c satisfy $c=(a+bi)^3-107i$, where $i^2=-1$. Find c.

Published problem and solutions · MAA AMC.

1985 AIME Problem 5: A six-period recurrence

Integers a_n satisfy $a_n=a_{n-1}-a_{n-2}$ for $n\geq 3$. The first 1492 terms sum to 1985 and the first 1985 terms sum to 1492. Find the sum of the first 2001 terms.

Published problem and solutions · MAA AMC.

1985 AIME Problem 10: Values of a floor sum

How many integers from 1 through 1000 occur as floor(2x)+floor(4x)+floor(6x)+floor(8x) for real x?

Published problem and solutions · MAA AMC.

1986 AIME Problem 1: A fourth-root quadratic

Find the sum of all real solutions of x^(1/4)=12/(7-x^(1/4)), with the principal fourth root.

Published problem and solutions · MAA AMC.

1986 AIME Problem 2: A product of four radical conjugates

Evaluate$(\sqrt{5}+\sqrt{6}+\sqrt{7})(\sqrt{5}+\sqrt{6}-\sqrt{7})(\sqrt{5}-\sqrt{6}+\sqrt{7})(-\sqrt{5}+\sqrt{6}+\sqrt{7})$.

Published problem and solutions · MAA AMC.

1986 AIME Problem 3: Tangent from a sum and a reciprocal sum

Given tanx+tany=25 and cotx+coty=30, find tan(x+y).

Published problem and solutions · MAA AMC.

1986 AIME Problem 4: A common-sum linear system

For i=1,2,3,4,5, let $x_1+\cdots+x_5+x_i$ equal 6,12,24,48,96 respectively. Find $3x_4+2x_5$.

Published problem and solutions · MAA AMC.

1986 AIME Problem 8: Logarithms of proper divisors

Let S be the sum of the base 10 logarithms of all positive proper divisors of 1,000,000. Find the integer nearest to S.

Published problem and solutions · MAA AMC.

1986 AIME Problem 10: Recovering a permuted numeral

The five digit permutations of a three-digit numeral other than its original ordering have sum 3194. Recover the original numeral, allowing leading zeros in rearrangements.

Published problem and solutions · MAA AMC.

1986 AIME Problem 11: A translated coefficient of a geometric polynomial

Write $1-x+x^2-\cdots-x^{17}$ as a polynomial $a_0+a_1y+\cdots+a_{17}y^{17}$ in y=x+1. Find a_2.

Published problem and solutions · MAA AMC.

1986 AIME Problem 14: A box from distances to a body diagonal

For a rectangular box, the distances from a body diagonal to the six edges it does not meet take the three values $2\sqrt{5}$,30/$\sqrt{13}$,15/$\sqrt{10}$. Find the box volume.

Published problem and solutions · MAA AMC.

1987 AIME Problem 4: A region bounded by absolute values

Find the area enclosed by|x-60|+|y|=|x|/4.

Published problem and solutions · MAA AMC.

1987 AIME Problem 5: A factored equation in two integer squares

Integers x,y satisfy $y^2+3x^2y^2=30x^2+517$. Find $3x^2y^2$.

Published problem and solutions · MAA AMC.

1987 AIME Problem 10: Walking against an escalator

Al walks down an upward-moving escalator, taking 150 walking steps. Bob walks up it, taking 75 walking steps. Al's walking rate is three times Bob's. How many steps are visible on the escalator?

Published problem and solutions · MAA AMC.

1987 AIME Problem 12: An integer just above a cube

Let m be the smallest integer with cube root n+r, where n is a positive integer and 0<r<1/1000. Find n.

Published problem and solutions · MAA AMC.

1987 AIME Problem 14: A telescoping Sophie Germain product

Evaluate the product over j=0,1,2,3,4 of$[((10+12j)^4+324)/((4+12j)^4+324)]$.

Published problem and solutions · MAA AMC.

1987 AIME Problem 15: Two orientations of an inscribed square

Triangle ABC is right-angled at C. One inscribed square has a vertex at C, sides along CA and CB, and its opposite vertex on AB; its area is 441. A second has one side on AB and its other two vertices on CA and CB; its area is 440. Find AC+CB.

Published problem and solutions · MAA AMC.

1988 AIME Problem 3: Nested logarithms with different bases

A real x satisfies $\log_2(\log_8 x)=\log_8(\log_2 x)$. Find$(\log_2 x)^2$.

Published problem and solutions · MAA AMC.

1988 AIME Problem 4: An absolute-value defect of nineteen

Real numbers $x_1,\ldots,x_n$ satisfy$|x_i|<1$ and $\Sigma|x_i|=19+|\Sigma x_i|$. Find the least possible n.

Published problem and solutions · MAA AMC.

1988 AIME Problem 6: A grid arithmetic in both directions

Index a5×5 grid by coordinates(i,j),0≤i,j≤4, increasing rightward and upward. Every row and column is arithmetic. Given T(0,0)=0,T(1,3)=74,T(4,2)=186,T(2,1)=103, and positive integers in all other cells, find T(3,4).

Published problem and solutions · MAA AMC.

1988 AIME Problem 8: A functional equation hiding the least common multiple

On positive integer pairs, f satisfies f(x,x)=x, f(x,y)=f(y,x), and (x+y)f(x,y)=y f(x,x+y). Find f(14,52).

Published problem and solutions · MAA AMC.

1988 AIME Problem 11: Mean lines pass through a centroid

A line is a mean line for complex points $w_1,\ldots,w_5$ if it contains $z_1,\ldots,z_5$ with $\Sigma(z_k-w_k)=0$. For w=(32+170i,-7+64i,-9+200i,1+27i,-14+43i), find the slope of the mean line whose y-intercept is 3.

Published problem and solutions · MAA AMC.

1988 AIME Problem 12: Three equal terminal pieces of cevians

Cevians AD,BE,CF of triangle ABC meet at an interior point P. Suppose PD=PE=PF=3, while AP=a,BP=b,CP=c and a+b+c=43. Find abc.

Published problem and solutions · MAA AMC.

1988 AIME Problem 13: Fibonacci remainders modulo a quadratic

Integers a,b satisfy x^2-x-1 | $ax^{17}+bx^{16}+1$ as polynomials. Find a.

Published problem and solutions · MAA AMC.

1988 AIME Problem 14: Reflecting a rectangular hyperbola

Reflect the curve xy=1 in the line y=2x. Write its equation as $12x^2+bxy+cy^2+d=0$. Find bc.

Published problem and solutions · MAA AMC.

1989 AIME Problem 1: A square hidden among four consecutive factors

Evaluate $\sqrt{(31\cdot 30\cdot 29\cdot 28+1)}$.

Published problem and solutions · MAA AMC.

1989 AIME Problem 3: A repeated decimal digit forced by divisibility

For a positive integer n and a decimal digit d, n/810=0.d25d25d25…. Find n.

Published problem and solutions · MAA AMC.

1989 AIME Problem 6: The first possible moving rendezvous

A and B are 100 units apart. Allie leaves A at speed 8 along a ray making a 60° angle with AB. Billie leaves B simultaneously at speed 7 along a chosen straight ray. What distance has Allie traveled at their earliest possible meeting?

Published problem and solutions · MAA AMC.

1989 AIME Problem 7: Squares of an arithmetic progression

An integer k is added to 36, 300, and 596. The three resulting numbers, in that order, are squares of three consecutive terms of an arithmetic progression. Find k.

Published problem and solutions · MAA AMC.

1989 AIME Problem 8: A quadratic sequence of linear forms

Real numbers $x_1,\ldots,x_7$ satisfy $\Sigma i^2x_i=1$, $\Sigma(i+1)^2x_i=12$, and $\Sigma(i+2)^2x_i=123$, with every sum over $i=1,\ldots,7$. Find $\Sigma(i+3)^2x_i$.

Published problem and solutions · MAA AMC.

1989 AIME Problem 14: Real integers written in a complex base

Let $\beta=-3+i$. Consider four-digit expressions $a_3\beta^3+a_2\beta^2+a_1\beta+a_0$ with decimal digits $0\le a_j\le 9$ and $a_3\neq 0$. Find the sum of all distinct real integers represented.

Published problem and solutions · MAA AMC.

1990 AIME Problem 2: A difference of conjugate radical cubes

Evaluate $(52+6\sqrt{43})^{(3/2)}-(52-6\sqrt{43})^{(3/2)}$.

Published problem and solutions · MAA AMC.

1990 AIME Problem 4: Three denominators with a common quadratic part

Find the positive real solution of $1/(x^2-10x-29)+1/(x^2-10x-45)-2/(x^2-10x-69)=0$.

Published problem and solutions · MAA AMC.

1990 AIME Problem 10: Products of two root-of-unity groups

Let $A=\{z : z^{18}=1\}$ and $B=\{w : w^{48}=1\}$. How many distinct complex numbers occur as zw with $z\in A$ and $w\in B$?

Published problem and solutions · MAA AMC.

1990 AIME Problem 15: A recurrence for weighted power sums

Real a,b,x,y satisfy $ax+by=3$, $ax^2+by^2=7$, $ax^3+by^3=16$, and $ax^4+by^4=42$. Find ax^5+by^5.

Published problem and solutions · MAA AMC.

1991 AIME Problem 1: Symmetric sums with an integer restriction

Positive integers x,y satisfy xy+x+y=71 and $x^2y+xy^2=880$. Find x^2+y^2.

Published problem and solutions · MAA AMC.

1991 AIME Problem 2: Parallel slices of a rectangle

In a 4 by 3 rectangle ABCD, divide AB and CB into 168 equal parts starting at A and C, and join corresponding interior division points. Repeat on AD and CD, then include diagonal AC. Find the total length of the resulting 335 parallel segments.

Published problem and solutions · MAA AMC.

1991 AIME Problem 3: The peak of a weighted binomial row

For $0 \le k \le 1000$, let $A_k = \binom{1000}{k}(0.2)^k$. Find the index k for which A_k is largest.

Published problem and solutions · MAA AMC.

1991 AIME Problem 4: Counting logarithm-sine intersections rigorously

How many real x satisfy $(\log_2 x)/5=\sin(5\pi x)$?

Published problem and solutions · MAA AMC.

1991 AIME Problem 6: Recovering a number from a floor sum

A real r satisfies $\Sigma_{k=19}{}^{91}$ floor(r+k/100)=546. Find floor(100r).

Published problem and solutions · MAA AMC.

1991 AIME Problem 7: A finite continued fraction and its fixed points

Let $f(t)=\sqrt{19+91/t}$. Find the square of the sum of the absolute values of all real solutions of x=f(f(f(f(f(x))))), with every intermediate expression defined.

Published problem and solutions · MAA AMC.

1991 AIME Problem 8: Integer roots via a shifted factorization

For how many real values a does $x^2+ax+6a=0$ have two integer roots, counted with multiplicity?

Published problem and solutions · MAA AMC.

1991 AIME Problem 9: A rational transformation of trigonometric sums

Suppose sec x+tan x=22/7. Write csc x+cot x=m/n in lowest terms with n>0. Find m+n.

Published problem and solutions · MAA AMC.

1991 AIME Problem 15: An attainable vector minimum that is integral

For each positive integer n, let S_n be the minimum of $\Sigma_{k=1}{}^{n}\sqrt{((2k-1)^2+a_k{}^2)}$ over positive real a_k with $\Sigma a_k=17$. Find the unique n for which S_n is an integer.

Published problem and solutions · MAA AMC.

1992 AIME Problem 3: A small improvement in a win ratio

Before a weekend a tennis player has won exactly half her matches. She then wins three of four matches, raising her win ratio above 0.503. What is the largest possible number of wins she had beforehand?

Published problem and solutions · MAA AMC.

1992 AIME Problem 4: Three consecutive binomial coefficients in a ratio

Rows of Pascal's triangle are numbered from row 0, containing only 1. In which row do three consecutive entries occur in the ratio $3 : 4 : 5$?

Published problem and solutions · MAA AMC.

1992 AIME Problem 8: A quadratic sequence from second differences

A real sequence satisfies $a_{n+2}-2a_{n+1}+a_n=1$ for every positive n, with $a_{19}=a_{92}=0$. Find a_1.

Published problem and solutions · MAA AMC.

1992 AIME Problem 10: A square outside two semicircles

Let A consist of nonzero complex z such that the real and imaginary parts of both z/40 and 40/conj(z) lie in [0,1]. Find the integer nearest the area of A.

Published problem and solutions · MAA AMC.

1992 AIME Problem 13: Maximum area on an Apollonius circle

Triangle ABC has AB=9 and BC:AC=40:41. Find its largest possible area.

Published problem and solutions · MAA AMC.

1992 AIME Problem 14: A universal product identity for cevian ratios

Cevians AA',BB',CC' of triangle ABC meet at an interior point O, with A',B',C' on the opposite sides. If $AO/OA'+BO/OB'+CO/OC'=92$, find the product of these three ratios.

Published problem and solutions · MAA AMC.

1993 AIME Problem 2: A forty-day spiral of square distances

A traveler goes east, north, west, south in a repeating four-day cycle. On day n he travels $n^2/2$ miles. How far is he from his starting point after 40 days?

Published problem and solutions · MAA AMC.

1993 AIME Problem 3: Recovering a total from overlapping averages

In a fishing contest the winner catches 15 fish. Exactly 9,5,7 contestants catch 0,1,2 fish respectively, and 5,2,1 catch 13,14,15 respectively. Those catching at least 3 average 6 fish; those catching at most 12 average 5. Find the total fish caught.

Published problem and solutions · MAA AMC.

1993 AIME Problem 5: Twenty translations of a polynomial

Let $P_0(x)=x^3+313x^2-77x-8$, and define $P_n(x)=P_{n-1}(x-n)$ for $n\geq 1$. Find the coefficient of x in $P_{20}(x)$.

Published problem and solutions · MAA AMC.

1993 AIME Problem 13: When a moving line of sight clears a building

Jenny and Kenny walk in the same direction at 1 and 3 feet per second on parallel paths 200 feet apart. A circular building of diameter 100 is centered midway between the paths. When it first blocks their view they are 200 feet apart. If visibility returns after t seconds, find numerator plus denominator of t in lowest terms.

Published problem and solutions · MAA AMC.

1993 AIME Problem 14: The least perimeter of an unstuck inscribed rectangle

A rectangle inscribed in a 6 by 8 rectangle has one vertex on each side. Call it unstuck if a nonzero sufficiently small rotation about its center keeps it inside the larger rectangle. The least possible perimeter is $\sqrt{N}$. Find N.

Published problem and solutions · MAA AMC.

1994 AIME Problem 2: A square between internally tangent circles

A circle of diameter PQ=10 is internally tangent at P to a circle of radius 20. A square ABCD has A,B on the larger circle, while CD is tangent to the smaller circle at Q and the smaller circle lies outside the square. If $AB=m+\sqrt{n}$, find m+n.

Published problem and solutions · MAA AMC.

1994 AIME Problem 3: A triangular particular solution

A real function satisfies $f(x)+f(x-1)=x^2$ for every real x, and f(19)=94. Find the remainder when f(94) is divided by 1000.

Published problem and solutions · MAA AMC.

1994 AIME Problem 4: A sum of binary logarithm floors

Find the positive integer n satisfying $\Sigma_{k=1}{}^{n}\ \text{floor}(\log_2 k)=1994$.

Published problem and solutions · MAA AMC.

1994 AIME Problem 8: An equilateral triangle from two heights

The points (0,0),(a,11),(b,37) form an equilateral triangle. Find ab.

Published problem and solutions · MAA AMC.

1994 AIME Problem 13: Reciprocal roots on a translated unit circle

The ten roots of $x^{10}+(13x-1)^{10}=0$ form five conjugate pairs $r_j,\text{conj}(r_j)$. Find $\Sigma_{j=1}{}^{5} 1/(r_j\,\text{conj}(r_j))$.

Published problem and solutions · MAA AMC.

1995 AIME Problem 1: Five shrinking squares with quarter overlaps

Start with a unit square. Each successive square has half the previous side length and is centered at its outward corner, with parallel sides; at every stage use the corner away from the earlier squares. For the first five squares, write their union area as m/n in lowest terms. Find m−n.

Published problem and solutions · MAA AMC.

1995 AIME Problem 2: A product of roots without solving them

Find the last three decimal digits of the product of all positive roots of $\sqrt{1995} \cdot x^{(\log_{1995} x)} = x^2$.

Published problem and solutions · MAA AMC.

1995 AIME Problem 5: A quartic coefficient from conjugate data

A monic quartic with real coefficients has four nonreal roots. The product of two roots is 13+i, and the sum of the other two is 3+4i. Find the coefficient of x^2.

Published problem and solutions · MAA AMC.

1995 AIME Problem 7: A symmetric sine-cosine expression

Given $(1+\sin t)(1+\cos t)=5/4$ and $(1-\sin t)(1-\cos t)=m/n-\sqrt{k}$, where k,m,n are positive integers and m/n is reduced, find k+m+n.

Published problem and solutions · MAA AMC.

1995 AIME Problem 9: An isosceles triangle with a tripled angle

In isosceles triangle ABC, AB=AC and altitude AM=11. A point D on AM satisfies AD=10 and $\angle BDC = 3\angle BAC$. If the triangle's perimeter is $a+\sqrt{b}$, find a+b.

Published problem and solutions · MAA AMC.

1995 AIME Problem 13: Grouping a sum by rounded fourth roots

Let f(n) be the integer nearest to the positive fourth root of n. Find $\Sigma_{k=1}{}^{1995} 1/f(k)$.

Published problem and solutions · MAA AMC.

1996 AIME Problem 1: The center of a three-by-three magic square

A 3×3 magic square has first row (x,19,96) and second-row first entry 1. Every row, column, and main diagonal has the same sum. Find x.

Published problem and solutions · MAA AMC.

1996 AIME Problem 3: Counting monomials after expansion

Find the least positive n for which $(xy-3x+7y-21)^n$ has at least 1996 nonzero monomials after collecting terms.

Published problem and solutions · MAA AMC.

1996 AIME Problem 5: A polynomial evaluated at its root sum

The roots of $x^3+3x^2+4x-11$ are a,b,c. A monic cubic with roots a+b,b+c,c+a has constant term t. Find t.

Published problem and solutions · MAA AMC.

1996 AIME Problem 10: An integer angle from a tangent identity

Find the smallest positive integer x satisfying $\tan(19x^\circ)=(\cos 96^\circ+\sin 96^\circ)/(\cos 96^\circ-\sin 96^\circ)$.

Published problem and solutions · MAA AMC.

1996 AIME Problem 11: Upper-half-plane roots of a reciprocal polynomial

Let P be the product of the roots of $z^6+z^4+z^3+z^2+1=0$ with positive imaginary part. Write $P=r(\cos\theta^\circ+i\sin\theta^\circ)$, $r>0$ and $0\le\theta<360$. Find θ.

Published problem and solutions · MAA AMC.

1996 AIME Problem 15: A parallelogram determined by doubled angles

Diagonals of parallelogram ABCD meet at O. Both $\angle$CAB and $\angle$DBC equal twice $\angle$DBA. If $\angle ACB=r\angle AOB$, find floor(1000r).

Published problem and solutions · MAA AMC.

1997 AIME Problem 3: Concatenation mistaken for multiplication

A two-digit integer x written immediately before a three-digit integer y forms a five-digit number equal to nine times xy. Find x+y.

Published problem and solutions · MAA AMC.

1997 AIME Problem 7: The midpoint time of a storm crossing

A car moves east at 2/3 mile per minute. A circular storm of radius 51 moves southeast at $\sqrt{2}/2$ mile per minute, initially centered 110 miles north of the car. If the car enters at t_1 and leaves at t_2, find $(t_1+t_2)/2$.

Published problem and solutions · MAA AMC.

1997 AIME Problem 9: Reducing a golden-ratio power

For positive a, the fractional parts of 1/a and a^2 are equal, and $2<a^2<3$. Find $a^{12}-144/a$.

Published problem and solutions · MAA AMC.

1997 AIME Problem 11: A reflected index in trigonometric sums

Let $x=(\Sigma_{n=1}{}^{44}\cos n^\circ)/(\Sigma_{n=1}{}^{44}\sin n^\circ)$. Find floor(100x).

Published problem and solutions · MAA AMC.

1997 AIME Problem 12: The missing value of a fractional-linear involution

Let f(x)=(ax+b)/(cx+d), where a,b,c,d are nonzero real numbers. Suppose f(19)=19, f(97)=97, and f(f(x))=x on its domain. Find the unique real number outside its range.

Published problem and solutions · MAA AMC.

1997 AIME Problem 13: A wire graph made of unit diagonals

Let f(t)=abs(abs(abs(t)−2)−1). The planar set S is defined by f(x)+f(y)=1. Its total length is a$\sqrt{}$b with b squarefree. Find a+b.

Published problem and solutions · MAA AMC.

1997 AIME Problem 14: A short arc probability

Two distinct 1997th roots of unity are chosen uniformly. Find numerator plus denominator for the chance that $|v+w| \geq \sqrt{2+\sqrt{3}}$.

Published problem and solutions · MAA AMC.

1997 AIME Problem 15: A rigorous maximum equilateral triangle in a rectangle

An equilateral triangle lies entirely inside a 10 by 11 rectangle. Its largest possible area is p$\sqrt{}$q−r, where q is squarefree. Find p+q+r.

Published problem and solutions · MAA AMC.

1998 AIME Problem 3: Factoring a graph into four half-lines

The graph y^2+2 xy+40|x|=400 encloses one bounded region. Find its area.

Published problem and solutions · MAA AMC.

1998 AIME Problem 5: A four-term cycle of signed triangular numbers

For positive k let A_k=k(k−1)cos(k(k−1)π/2)/2. Find $|A_{19}+A_{20}+\cdots+A_{98}|$.

Published problem and solutions · MAA AMC.

1998 AIME Problem 8: The longest subtractive recurrence

A sequence begins 1000,x and each later term is the term two places earlier minus the preceding term. Stop when the first negative term appears, including it. Which positive integer x makes the sequence longest?

Published problem and solutions · MAA AMC.

1998 AIME Problem 12: A golden-ratio configuration in a medial triangle

In equilateral triangle ABC, D,E,F are the midpoints of BC,CA,AB. Points P∈DE,Q∈EF,R∈FD satisfy P∈CQ, Q∈AR, and R∈BP. If [ABC]/[PQR]=a+b$\sqrt{}$c with c squarefree, find $a^2+b^2+c^2$.

Published problem and solutions · MAA AMC.

1998 AIME Problem 13: Complex weights over subsets

For each nonempty subset of {1,...,n}, list its elements increasingly and weight the jth by i^j. Let S_n sum these weighted subset sums. Given S_8=−176−64i and S_9=p+qi, find |p|+|q|.

Published problem and solutions · MAA AMC.

1998 AIME Problem 14: A sharp bound for a box volume ratio

Positive integers m≤n≤p satisfy 2 mnp=(m+2)(n+2)(p+2). Find the largest possible p.

Published problem and solutions · MAA AMC.

1999 AIME Problem 3: A quadratic square condition with four solutions

Find the sum of all positive integers n for which $n^2-19n+99$ is a perfect square.

Published problem and solutions · MAA AMC.

1999 AIME Problem 6: Square roots turn a quadrilateral into an annular sector

Apply (x,y)↦($\sqrt{x}$,$\sqrt{y}$) to the region enclosed by (900,300),(1800,600),(600,1800),(300,900). If its image has area k, find floor(k).

Published problem and solutions · MAA AMC.

1999 AIME Problem 9: An equidistant complex multiplier

For positive real a,b, the map f(z)=(a+bi)z sends every complex z to a point equidistant from z and 0. Given |a+bi|=8 and b^2=m/n in lowest terms, find m+n.

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1999 AIME Problem 11: A telescoping arithmetic-angle sine sum

Angles are in degrees. Given $\Sigma_{k=1}{}^{35}\sin(5k°)=\tan((m/n)°)$, where m,n are coprime positive integers and m/n<90, find m+n.

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1999 AIME Problem 12: A triangle from inradius and two tangent lengths

The incircle of triangle ABC has radius 21 and touches AB at P, with AP=23 and PB=27. Find the triangle's perimeter.

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2000 AIME I Problem 3: Adjacent binomial coefficients

For coprime positive integers a, b, the coefficients of x^2 and x^3 in $(ax+b)^{2000}$ agree. Determine $a+b$.

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2000 AIME I Problem 4: Nine-square dissection as a linear system

A rectangle is tiled by nine squares with side lengths $a_1 < \cdots < a_9$. Their edge relations are $a_3 = a_1 + a_2$, $a_4 = a_1 + a_3$, $a_5 = a_3 + a_4$, $a_6 = a_4 + a_5$, $a_7 = a_2 + a_3 + a_5$, $a_8 = a_2 + a_7$, $a_9 = a_1 + a_4 + a_6$, and $a_6 + a_9 = a_7 + a_8$. Its width is $a_8 + a_9$ and height is $a_6 + a_9$. If these dimensions are coprime integers, find its perimeter.

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2000 AIME I Problem 6: A gap between means

How many integer pairs (x, y) satisfy $0 < x < y < 10^6$ and $\frac{x+y}{2} - \sqrt{xy} = 2$?

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2000 AIME I Problem 7: A reciprocal cyclic system

Positive reals satisfy $xyz = 1$, $x + 1/z = 5$, and $y + 1/x = 29$. Express $z + 1/y = m/n$ in lowest terms and find $m + n$.

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2000 AIME I Problem 9: Shifted logarithms factor a system

Positive reals satisfy $\log(2000xy) - \log x \log y = 4$, $\log(2yz) - \log y \log z = 1$, and $\log(zx) - \log z \log x = 0$, with base-10 logarithms. Find the sum of all possible values of y.

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2000 AIME I Problem 10: A hundred variables and one sum

Reals $x_1, \ldots, x_{100}$ satisfy $x_k = \sum_{j \neq k} x_j - k$ for each k. If $x_{50} = m/n$ in lowest terms with $n > 0$, find $m + n$.

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2000 AIME I Problem 12: Reflection orbits

A real function satisfies $f(x) = f(398 - x) = f(2158 - x) = f(3214 - x)$ for every real x. What is the largest number of different values among $f(0), \ldots, f(999)$?

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2000 AIME I Problem 13: Fast highways and a slower prairie

Two perpendicular highways cross at the fire station. A truck travels at 50 miles per hour on either highway and 14 miles per hour elsewhere. The area it can reach in six minutes is m/n square miles in lowest terms. Find $m + n$.

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2000 AIME II Problem 1: Reciprocal logarithms share a base

Write $\frac{2}{\log_4(2000^6)} + \frac{3}{\log_5(2000^6)} = m/n$ in lowest terms. Find $m + n$.

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2000 AIME II Problem 7: Half a binomial row

Define N by $\sum_{k=2}^{9} 1/(k!(19-k)!) = N/18!$. Find the greatest integer strictly smaller than $N/100$.

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2000 AIME II Problem 9: Reciprocal complex powers

If $z \in \mathbb{C}$ satisfies $z + 1/z = 2\cos 3^\circ$, find the least integer strictly larger than $z^{2000} + z^{-2000}$.

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2000 AIME II Problem 10: Radius from four tangent lengths

A circle inscribed in convex quadrilateral $ABCD$ touches AB at P and CD at Q. If $AP = 19$, $PB = 26$, $CQ = 37$, and $QD = 23$, find the square of its radius.

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2000 AIME II Problem 13: A hidden geometric-series factor

The polynomial $2000x^6 + 100x^5 + 10x^3 + x - 2$ has a real root $(m + \sqrt{n})/r$, where m, n, r are integers, $r > 0$, and $\gcd(m, r) = 1$. Find $m + n + r$.

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2000 AIME II Problem 14: Borrowing in factorial base

For $N = 16! - 32! + 48! - \cdots - 1984! + 2000!$, let $N = \sum_{i=1}^{j} f_i i!$ be its factorial-base expansion, where $0 \le f_i \le i$ and $f_j > 0$. Find $\sum_{i=1}^{j} (-1)^{i+1} f_i$.

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2000 AIME II Problem 15: Reflection completes a trigonometric telescope

Find the least positive integer n such that $\sum_{k=0}^{44} \frac{1}{\sin(45+2k)^\circ \sin(46+2k)^\circ} = 1/\sin n^\circ$.

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2001 AIME I Problem 2: Two changes to a mean

A finite set of distinct reals has mean M. Adjoining 1 lowers its mean by 13, while adjoining 2001 raises it by 27. Find M.

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2001 AIME I Problem 3: A polynomial symmetric about one quarter

Find the sum of all complex roots of $x^{2001} + (1/2 - x)^{2001} = 0$, assuming all roots are simple.

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2001 AIME I Problem 5: Equilateral triangle in an ellipse

An equilateral triangle inscribed in $x^2 + 4y^2 = 4$ has vertex $(0, 1)$ and an altitude on the y-axis. If its squared side length is reduced m/n, find $m + n$.

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2001 AIME I Problem 9: Area from symmetric side fractions

In triangle ABC with sides $13, 15, 17$, points $D \in AB, E \in BC, F \in CA$ satisfy $AD/AB = p$, $BE/BC = q$, $CF/CA = r$, where $p, q, r > 0$, $p + q + r = 2/3$, and $p^2 + q^2 + r^2 = 2/5$. If $[DEF]/[ABC] = m/n$ in lowest terms, find $m + n$.

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2001 AIME II Problem 3: A fourth-order recurrence hides a sign flip

Let $x_1 = 211, x_2 = 375, x_3 = 420, x_4 = 523$ and $x_n = x_{n-1} - x_{n-2} + x_{n-3} - x_{n-4}$ for $n \geq 5$. Find $x_{531} + x_{753} + x_{975}$.

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2001 AIME II Problem 4: A prescribed midpoint between two lines

Points P, Q lie on $8y = 15x$ and $10y = 3x$, respectively, and have midpoint $(8, 6)$. If $PQ = m/n$ in lowest terms, find $m + n$.

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2001 AIME II Problem 8: The first crossing of a self-similar tent function

For positive x, a function satisfies $f(3x) = 3f(x)$ and $f(x) = 1 - |x - 2|$ on $1 \leq x \leq 3$. Find the smallest positive x with $f(x) = f(2001)$.

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2001 AIME II Problem 14: Unit roots of a sparse polynomial

List the unit-modulus roots of $z^{28} - z^8 - 1 = 0$ by arguments $0 \leq \theta_1 < \cdots < \theta_{2n} < 360^\circ$. Find $\theta_2 + \theta_4 + \cdots + \theta_{2n}$.

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2002 AIME I Problem 4: A telescope followed by a prime-square factorization

For $a_k = 1/(k^2 + k)$, positive integers $m < n$ satisfy $\sum_{k=m}^{n-1} a_k = 1/29$. Find $m + n$.

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2002 AIME I Problem 6: Products of exponential solutions without solving them

The system $\log_{225} x + \log_{64} y = 4$, $\log_x 225 - \log_y 64 = 1$ has solutions $(x_1, y_1), (x_2, y_2)$. Find $\log_{30}(x_1 y_1 x_2 y_2)$.

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2002 AIME I Problem 7: Three fractional digits of an enormous power

Find the first three digits after the decimal point of $(10^{2002}+1)^{10/7}$.

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2002 AIME I Problem 8: The first repeated Fibonacci-type endpoint

Consider nondecreasing positive integer sequences satisfying $a_n = a_{n-1} + a_{n-2}$ for $n > 2$. Find the smallest value of a_9 attained by at least two such sequences.

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2002 AIME I Problem 12: A Möbius transformation of order three

Let $F(z) = (z+i)/(z-i)$ and $z_{n+1} = F(z_n)$, starting from $z_0 = 1/137 + i$. If $z_{2002} = a + bi$, find $a + b$.

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2002 AIME II Problem 3: An integral geometric progression with a square gap

Positive integers $a < b < c$ form a geometric progression and satisfy $\log_6 a + \log_6 b + \log_6 c = 6$. If $b - a$ is a perfect square, find $a + b + c$.

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2002 AIME II Problem 6: A reciprocal-square telescope with a rounding bound

Find the integer nearest to $1000 \sum_{n=3}^{10000} 1/(n^2-4)$.

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2002 AIME II Problem 8: The first missing integer quotient

Find the least positive integer k never equal to $\lfloor 2002/n \rfloor$ for an integer $n \neq 0$.

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2002 AIME II Problem 10: Equal sine values in degrees and radians

The two smallest positive real numbers for which $\sin(x^\circ) = \sin x$ are $m\pi/(n-\pi)$ and $p\pi/(q+\pi)$, with positive integer parameters. Find $m+n+p+q$.

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2002 AIME II Problem 11: Two geometric series with a shared second term

Two distinct convergent real geometric series each sum to 1 and have the same second term. One has third term $1/8$. Write the common second term as $(\sqrt{m}-n)/p$, with square-free m, and find $100m + 10n + p$.

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2002 AIME II Problem 14: A circle centered on a leg of a right triangle

Right triangle APM has right angle at A and perimeter 152. A circle of radius 19, centered at $O \in AP$, is tangent to AM and PM. Write $OP = m/n$ in lowest terms and find $m + n$.

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2002 AIME II Problem 15: Two tangent circles through one point

Two circles meet in two points, including $(9, 6)$. Both are tangent to the x-axis and to $y = mx$, where $m > 0$, and their radii have product 68. Write $m = a\sqrt{b}/c$ in lowest radical form and find $a + b + c$.

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2003 AIME I Problem 2: Alternating concentric annuli

Concentric circles have radii $1, 2, \ldots, 100$. Color the central disk red and alternate red and green on successive annuli. If the green area divided by the radius-100 disk area is reduced m/n, find $m + n$.

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2003 AIME I Problem 4: A logarithmic trigonometric invariant

Given $\log_{10} \sin x + \log_{10} \cos x = -1$ and $\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$, find n.

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2003 AIME I Problem 8: An arithmetic triple followed by a geometric triple

Four increasing positive integers have their first three in arithmetic progression and their last three in geometric progression. The last exceeds the first by 30. Find their sum.

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2003 AIME I Problem 11: Triangle inequalities as interval probability

Choose x uniformly from $(0, 90^\circ)$. The probability that $\sin^2 x, \cos^2 x, \sin x \cos x$ fail to form a triangle is d/n, where d is the degree measure of $\arctan m$ and $m + n < 1000$. Find $m + n$.

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2003 AIME I Problem 12: Equal opposite angles cancel two cosine laws

Convex quadrilateral $ABCD$ has $\angle A = \angle C$, $AB = CD = 180$, $AD \neq BC$, and perimeter 640. Find $\lfloor 1000 \cos A \rfloor$.

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2003 AIME II Problem 8: Products of arithmetic sequences are quadratic

A sequence begins $1440, 1716, 1848, \ldots$ and is formed by multiplying corresponding terms of two arithmetic sequences. Find its eighth term.

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2003 AIME II Problem 9: Reduce a polynomial before summing it over roots

Let $P(x) = x^6 - x^5 - x^3 - x^2 - x$ and $Q(x) = x^4 - x^3 - x^2 - 1$. If $z_1, \ldots, z_4$ are the roots of Q, counted with multiplicity, find $\sum_{j=1}^{4} P(z_j)$.

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2003 AIME II Problem 14: An equilateral hexagon with six prescribed heights

A convex equilateral hexagon ABCDEF has $A = (0,0)$, $B = (b,2)$, $\angle FAB = 120^\circ$, and opposite sides parallel. Its vertex y-coordinates are exactly $0, 2, 4, 6, 8, 10$. Write its area as $m\sqrt{n}$ with square-free n, and find $m + n$.

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2003 AIME II Problem 15: Triangular coefficients encode a squared geometric sum

Let $P(x) = 24x^{24} + \sum_{j=1}^{23}(24-j)(x^{24-j} + x^{24+j})$. For its distinct roots z, write $\sum_z |\operatorname{Im}(z^2)| = m + n\sqrt{p}$ with square-free p. Find $m + n + p$.

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2004 AIME I Problem 2: Consecutive sets from their means

A set A of m consecutive integers sums to $2m$. A set B of $2m$ consecutive integers sums to m. Their greatest elements differ in absolute value by 99. Find m.

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2004 AIME I Problem 5: Strict daily rates with a larger aggregate score

A contestant attempts 500 points over two days and scores a positive integer each day. Its first-day success rate is below 160/300, its second-day rate below 140/200, and it does not attempt 300 points on day one. Find m+n for the largest possible aggregate rate m/n in lowest terms.

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2004 AIME I Problem 7: Pair contributions to a coefficient

Find the absolute value of the coefficient of x^2 in $\prod_{k=1}^{15}(1 + (-1)^k kx)$.

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2004 AIME I Problem 11: Equal painted-area and volume ratios

A cone of height 4 and radius 3 is painted on its entire exterior, including its base. A parallel cut separates a smaller cone C and a frustum F. Their painted-area ratio C:F equals their volume ratio C:F, namely reduced m/n. Find m+n.

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2004 AIME I Problem 12: Area selected by even logarithmic floors

Let S consist of $(x, y) \in (0, 1]^2$ for which both $\lfloor \log_2(1/x) \rfloor$ and $\lfloor \log_5(1/y) \rfloor$ are even. If S has area reduced m/n, find m+n.

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2004 AIME I Problem 13: Two interlaced families of roots of unity

Write the 34 roots of $(1 + x + \cdots + x^{17})^2 - x^{17}$ as $r_k e^{2\pi i a_k}$, where $r_k > 0$ and $0 < a_1 \leq \cdots \leq a_{34} < 1$. If $a_1 + \cdots + a_5 = m/n$ in lowest terms, find m+n.

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2004 AIME II Problem 5: Recovering a delayed work schedule

A job was planned for 1000 equally productive workers. Its first quarter used all 1000, its second quarter used 900, and its third used 800. Find the fewest additional workers to hire for the last quarter so the original deadline is met.

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2004 AIME II Problem 6: Integer shares in a three-stage distribution

Three monkeys successively take portions of a banana pile. They keep respectively $3/4, 1/4, 1/12$ of their portions and split each remainder equally between the other two. Every transfer is integral, the last monkey empties the pile, and final holdings have ratio $3 : 2 : 1$. Find the smallest initial pile.

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2004 AIME II Problem 9: Alternating arithmetic and geometric triples

A positive integer sequence starts at $a_1 = 1$. Every triple $a_{2j-1}, a_{2j}, a_{2j+1}$ is geometric, and every triple $a_{2j}, a_{2j+1}, a_{2j+2}$ is arithmetic. Given $a_9 + a_{10} = 646$, let a_n be its greatest term below 1000. Find $n + a_n$.

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2004 AIME II Problem 12: A circle between four vertex circles

Isosceles trapezoid ABCD has $AB = 6, CD = 4, AD = BC = 5$. Circles centered at A,B have radius 3, and those at C,D have radius 2. A circle contained in the trapezoid is tangent to all four. Write its radius as $(-k + m\sqrt{n})/p$ with square-free n and $\gcd(k, p) = 1$, and find $k + m + n + p$.

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2005 AIME I Problem 4: A rectangular formation five above a square

A band has five more members than a square formation can hold, and exactly fills a rectangle with seven more rows than columns. Find its greatest possible size.

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2005 AIME I Problem 6: Complete a fourth power before separating roots

Let P be the product of the nonreal roots of $x^4 - 4x^3 + 6x^2 - 4x = 2005$. Find $\lfloor P \rfloor$.

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2005 AIME I Problem 8: A sum of exponential roots from a cubic product

The equation $2^{333x-2} + 2^{111x+2} = 2^{222x+1} + 1$ has three real roots whose sum is reduced m/n. Find m+n.

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2005 AIME I Problem 10: Move along a median to prescribe area

Triangle ABC has area 70, with $B = (12, 19)$ and $C = (23, 20)$. Its median from $A = (p, q)$ to BC has slope -5. Find the largest p+q.

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2005 AIME I Problem 11: Largest semicircle in a square

A semicircular region lies inside a square of side 8. Its largest possible diameter is $m - \sqrt{n}$ for positive integers m,n. Find m+n.

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2005 AIME I Problem 14: Constructing a square by perpendicular normal coordinates

A unique square has the points $A = (0, 12)$, $B = (10, 9)$, $C = (8, 0)$, $D = (-4, 7)$ on four different sides. If its area is K, find the remainder of 10K modulo 1000.

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2005 AIME I Problem 15: An incircle trisecting a median

Triangle ABC has $BC = 20$. Its incircle meets median AD at the two trisection points. If its area is $m\sqrt{n}$ with square-free n, find m+n.

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2005 AIME II Problem 1: A binomial ratio determines deck size

A deck with $n \geq 6$ distinct cards has six times as many six-card subsets as three-card subsets. Find n.

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2005 AIME II Problem 3: A geometric series and its squared terms

An infinite geometric series sums to 2005. Squaring every term produces a series whose sum is 20050. If the original common ratio is reduced m/n, find m+n.

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2005 AIME II Problem 5: Reciprocal logarithms force powers

Count ordered integer pairs (a, b) with $2 \leq a, b \leq 2005$ satisfying $\log_a b + 6 \log_b a = 5$.

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2005 AIME II Problem 7: A dyadic radical product telescopes

Let $x = 4/((\sqrt{5}+1)(\sqrt[4]{5}+1)(\sqrt[8]{5}+1)(\sqrt[16]{5}+1))$. Find $(x+1)^{48}$.

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2005 AIME II Problem 9: A swapped sine-cosine power identity

For how many positive integers $n \leq 1000$ does $(\sin t + i\cos t)^n = \sin(nt) + i\cos(nt)$ hold for every real t?

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2005 AIME II Problem 11: Adjacent products linearize a nonlinear recurrence

A real sequence satisfies $a_0 = 37, a_1 = 72, a_m = 0$ and $a_{k+1} = a_{k-1} - 3/a_k$ for $1 \leq k < m$. Find the positive integer m.

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2005 AIME II Problem 12: A forty-five-degree view of a segment

In square ABCD of side 900 and center O, points E,F lie on AB in that order, with $AE < BF$, $EF = 400$, and $\angle EOF = 45^\circ$. If $BF = p + q\sqrt{r}$ with square-free r, find p+q+r.

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2005 AIME II Problem 13: Integer polynomial values constrain integer roots

An integer-coefficient polynomial satisfies $P(17) = 10$, $P(24) = 17$, and $P(n) = n + 3$ for two distinct integers n_1, n_2. Find $n_1 n_2$.

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2005 AIME II Problem 15: Tangent-circle centers form an ellipse

Circles w_1, w_2 have equations $x^2 + y^2 + 10x - 24y - 87 = 0$ and $x^2 + y^2 - 10x - 24y + 153 = 0$. Let m be the least positive slope of a line through the origin containing the center of a circle internally tangent to w_1 and externally tangent to w_2. If $m^2 = p/q$ in lowest terms, find p+q.

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2006 AIME I Problem 5: Recognizing a squared radical sum

Positive integers a, b, c satisfy $a\sqrt{2} + b\sqrt{3} + c\sqrt{5} = \sqrt{2006 + 104\sqrt{6} + 468\sqrt{10} + 144\sqrt{15}}$. Find abc.

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2006 AIME I Problem 6: Uniform digit contributions

Sum all repeating decimals $0.\overline{abc}$ with three distinct decimal digits, allowing $a = 0$.

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2006 AIME I Problem 7: Alternating strips in an angle

An angle has its vertex between parallel lines L_0, L_1 in an equally spaced family. Both rays cross L_1, $L_2, \ldots$. Let A be its triangular part before L_1 and let B, C, D be its portions between L_2, L_3, between L_4, L_5, and between L_6, L_7, respectively. If $[C]/[B] = 11/5$, find $[D]/[A]$.

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2006 AIME I Problem 9: Integral geometric progressions and logarithms

A geometric progression begins with positive integer a and has positive integer ratio r. If $\sum_{k=0}^{11} \log_8(ar^k) = 2006$, how many pairs (a, r) exist?

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2006 AIME I Problem 12: Factoring a trigonometric cubic

Find the sum of all degree measures x in $(100, 200)$ satisfying $\cos^3(3x) + \cos^3(5x) = 8\cos^3(4x)\cos^3 x$.

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2006 AIME I Problem 15: A telescoping potential

Start with $x_0 = 0$ and choose x_k subject to $|x_k| = |x_{k-1} + 3|$. Find the minimum of $|x_1 + \cdots + x_{2006}|$.

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2006 AIME II Problem 2: Logarithms as triangle sides

For how many positive integers n do $\log_{10} 12$, $\log_{10} 75$, and $\log_{10} n$ form the sides of a nondegenerate triangle?

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2006 AIME II Problem 5: Opposite-face bias

A die assigns probability $1/6 + t$ to a face, $1/6 - t$ to its opposite, and $1/6$ to the four others; opposite labels sum to seven. Two independent such dice sum to seven with probability $47/288$. Find numerator plus denominator of the larger face probability.

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2006 AIME II Problem 6: An axis-parallel square in a right triangle

In unit square $ABCD$, points $E \in BC$ and $F \in CD$ make AEF equilateral. A smaller square lies inside $ABCD$, has vertex B, parallel sides, and its opposite vertex on AE. Its side is $(a - \sqrt{b})/c$ in reduced form with squarefree b. Find $a + b + c$.

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2006 AIME II Problem 9: Intersecting internal common tangents

Circles with centers $(0, 0), (12, 0), (24, 0)$ have radii $1, 2, 4$. The first two have a common internal tangent of positive slope; the last two have one of negative slope. Their intersection has x-coordinate $p - q\sqrt{r}$ in reduced squarefree form. Find $p + q + r$.

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2006 AIME II Problem 11: A telescoping tribonacci sum

Let $a_1 = a_2 = a_3 = 1$ and $a_{n+3} = a_{n+2} + a_{n+1} + a_n$. Given $a_{28} = 6090307$, $a_{29} = 11201821$, and $a_{30} = 20603361$, find $\sum_{k=1}^{28} a_k$ modulo 1000.

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2006 AIME II Problem 15: A radical system encoding triangle altitudes

Real numbers satisfy $x = \sqrt{y^2 - 1/16} + \sqrt{z^2 - 1/16}$, $y = \sqrt{z^2 - 1/25} + \sqrt{x^2 - 1/25}$, and $z = \sqrt{x^2 - 1/36} + \sqrt{y^2 - 1/36}$. If $x + y + z = m/\sqrt{n}$ with squarefree n and positive integers m, n, find $m + n$.

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2007 AIME I Problem 2: Three moving positions

A length-100 walkway moves at 6 feet per second. Al stands on it from time 0. Bob enters at time 2 and walks forward at 4 feet per second relative to it. Cy starts beside it at time 4, walking at 8 feet per second. When one is midway between the others, how far from the start is that person?

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2007 AIME I Problem 3: Equal imaginary parts

For $z = 9 + bi$ with $b > 0$, the numbers z^2 and z^3 have equal imaginary parts. Find b.

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2007 AIME I Problem 4: The next planetary alignment

Three coplanar planets travel uniformly in circular orbits in the same direction, with periods $60, 84, 140$ years. They and the central star are initially collinear. After how many years will this first recur?

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2007 AIME I Problem 7: A floor-ceiling logarithm indicator

Find $\sum_{k=1}^{1000} k(\lceil \log_{\sqrt{2}} k \rceil - \lfloor \log_{\sqrt{2}} k \rfloor)$ modulo 1000.

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2007 AIME I Problem 8: Two quadratic factors of a cubic

For what largest real k can both $x^2 + (k-29)x - k$ and $2x^2 + (2k-43)x + k$ divide a cubic polynomial?

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2007 AIME I Problem 12: Overlap of two rotated isosceles triangles

Let $A = (0,0)$, $B = (20,0)$, and let C lie in the first quadrant with $AC = BC$ and $\angle BAC = 75^\circ$. Rotate the triangle counterclockwise until C reaches the positive vertical axis. If the common area is $p\sqrt{2} + q\sqrt{3} + r\sqrt{6} + s$, find $(p - q + r - s)/2$.

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2007 AIME I Problem 14: Linearizing a nonlinear recurrence

Let $a_0 = a_1 = 3$ and $a_{n+1}a_{n-1} = a_n^2 + 2007$ for $n \geq 1$. Find $\lfloor (a_{2006}^2 + a_{2007}^2)/(a_{2006}a_{2007}) \rfloor$.

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2007 AIME I Problem 15: An inscribed sixty-degree triangle

In equilateral triangle ABC, points $F \in AB$, $D \in BC$, and $E \in CA$ satisfy $AF = 5$, $CD = 2$, $\angle DEF = 60^\circ$, and $[DEF] = 14\sqrt{3}$. The two possible side lengths are $p \pm q\sqrt{r}$ with rational p, q and squarefree integer r. Find r.

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2007 AIME II Problem 4: Worker-hours as an additive resource

Identical workers produce two products at fixed per-item labor costs. In one hour, 100 workers make 300 of the first and 200 of the second. In two hours, 60 workers make 240 and 300. In three hours, 50 workers make 150 of the first and m of the second. Find m.

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2007 AIME II Problem 7: Multiples inside a cube-root interval

Exactly 70 positive integers n satisfy both $\lfloor \sqrt[3]{n} \rfloor = k$ and $k \mid n$ for some integer k. Find the largest possible value of n/k among them.

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2007 AIME II Problem 8: Optimizing a rectangular grid

Draw full lines parallel to the sides of a 4×5 sheet, with total drawn length 2007. Basic rectangles must have four drawn sides and no drawn line inside. Find the maximal number modulo 1000.

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2007 AIME II Problem 11: Rolling over a smaller cylinder

A radius-72 cylinder rolls along a plane, over a fixed parallel radius-24 cylinder, and back onto the plane without slipping. After exactly one complete rotation, its horizontal displacement is $a\pi + b\sqrt{c}$, with integer coefficients and squarefree c. Find $a + b + c$.

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2007 AIME II Problem 12: The largest term controls a geometric sum

An increasing geometric sequence consists of integral powers of 3. It satisfies $\sum_{j=0}^{7} \log_3 x_j = 308$ and $56 \leq \log_3(\sum_{j=0}^{7} x_j) \leq 57$. Find $\log_3 x_{14}$.

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2007 AIME II Problem 14: A polynomial identity controlled by its roots

A real polynomial satisfies $f(0) = 1$, $f(2) + f(3) = 125$, and $f(x)f(2x^2) = f(2x^3 + x)$ identically. Find $f(5)$.

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2008 AIME I Problem 1: Changing a percentage by adding students

Initially a party is 60% girls, and 40% of attendees like dancing. Twenty more boys arrive, all liking dancing, and the party becomes 58% girls. How many attendees now like dancing?

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2008 AIME I Problem 3: Integer speeds from two distance totals

Biking, jogging, and swimming speeds b, j, s are positive integers. A two-hour bike ride, three-hour jog, and four-hour swim cover 74 km; four hours biking, two jogging, and three swimming cover 91 km. Find $b^2 + j^2 + s^2$.

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2008 AIME I Problem 4: A shifted difference of squares

Positive integers x, y satisfy $x^2 + 84x + 2008 = y^2$. Find $x + y$.

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2008 AIME I Problem 8: An arctangent sum through complex multiplication

Find positive integer n for which $\arctan(1/3) + \arctan(1/4) + \arctan(1/5) + \arctan(1/n) = \pi/4$.

Published problem and solutions · MAA AMC.

2008 AIME I Problem 10: A trapezoid forced by equality bounds

An isosceles trapezoid $ABCD$ has longer base $AD \parallel BC$, base angles 60°, and diagonals $10\sqrt{21}$. A point E satisfies $EA = 10\sqrt{7}$, $ED = 30\sqrt{7}$. If F is the perpendicular projection of C onto AD, write $EF = m\sqrt{n}$ in squarefree form and find $m + n$.

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2008 AIME I Problem 12: Maximum traffic through a fixed point

All cars are 4 meters long and travel at common speed $v > 0$ km/h. The gap behind each car is exactly $4\lceil v/15 \rceil$ meters. A roadside detector counts passing cars for one hour. Over all speeds and initial placements, let M be the largest possible whole count. Find the integer quotient $M/10$.

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2008 AIME I Problem 13: A common zero of a two-dimensional cubic family

Every polynomial $p(x, y)$ of total degree at most three under consideration vanishes at $(0, 0), (1, 0), (-1, 0), (0, 1), (0, -1), (1, 1), (1, -1), (2, 2)$. Find the positive nonintegral common zero $(a/c, b/c)$, with positive integers a, b, c, $\gcd(a, c) = 1$, $c > 1$, by reporting $a + b + c$.

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2008 AIME I Problem 14: Maximizing a tangent projection distance

Circle ω has diameter $AB = 18$. A point C lies beyond A on ray BA, and CT is tangent to the circle. Let P be the perpendicular projection of A onto CT. If m is the largest possible BP, find m^2.

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2008 AIME II Problem 1: Paired alternating squares

Find the remainder modulo 1000 of $100^2 + 99^2 - 98^2 - 97^2 + 96^2 + 95^2 - \cdots + 4^2 + 3^2 - 2^2 - 1^2$.

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2008 AIME II Problem 2: Equal arrival times with different rest schedules

Two cyclists start together on a 50-mile trip. The faster rider rests five minutes after each mile, and the slower rider, traveling at three-quarters the faster speed, rests five minutes after each two miles. They reach the finish together. How many minutes elapse?

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2008 AIME II Problem 3: Ten slices from a rectangular block

A $10 \times 13 \times 14$ rectangular solid loses ten slices, each one unit thick and parallel to a face of the remaining block. What is the largest possible remaining volume?

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2008 AIME II Problem 6: A factorial recurrence in disguise

Sequences satisfy $a_0 = a_1 = 1$, $b_0 = 1$, $b_1 = 3$, and both obey $u_n = u_{n-1} + u_{n-1}^2/u_{n-2}$ for $n \geq 2$. Find b_{32}/a_{32}.

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2008 AIME II Problem 7: Cubes of pairwise root sums

Let r, s, t be the roots, with multiplicity, of $8x^3 + 1001x + 2008 = 0$. Find $(r+s)^3 + (s+t)^3 + (t+r)^3$.

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2008 AIME II Problem 8: A trigonometric telescope and divisibility

Put $a = \pi/2008$. Find the smallest positive integer n for which $2\sum_{k=1}^{n} \cos(k^2 a)\sin(ka)$ is an integer.

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2008 AIME II Problem 9: An affine rotation with period eight

Start at $(5, 0)$. A move rotates the point counterclockwise by 45° about the origin and then translates it ten units right. After 150 moves the point is (p, q). Find $\lfloor |p| + |q| \rfloor$.

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2008 AIME II Problem 11: Two tangent circles along an isosceles base

Triangle ABC has $AB = AC = 100$, $BC = 56$. A radius-16 circle touches AC and BC. Another circle lies entirely inside the triangle, touches AB and BC, and is externally tangent to the first circle. Its radius is $m - n\sqrt{k}$ with squarefree k. Find $m + nk$.

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2008 AIME II Problem 13: Inverting the exterior of a regular hexagon

A regular hexagon centered at 0 in the complex plane has opposite sides one unit apart, including a pair parallel to the imaginary axis. Let $S = \{1/z : z \text{ is outside the hexagon}\}$. If its area is $a\pi + \sqrt{b}$, find $a + b$.

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2008 AIME II Problem 14: An equilateral constraint inside a rectangle

For $a \geq b > 0$, suppose $a^2 + y^2 = b^2 + x^2 = (a - x)^2 + (b - y)^2$ has a solution with $0 \leq x < a$, $0 \leq y < b$. Let ρ be the largest possible a/b. If $\rho^2 = m/n$ in lowest terms, find $m + n$.

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2009 AIME I Problem 1: Extremal three-digit geometric progressions

A three-digit integer is called geometric when its three distinct digits, in order, form a geometric progression. Find the largest such integer minus the smallest.

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2009 AIME I Problem 2: A complex quotient with fixed imaginary part

A complex number z has imaginary part 164 and satisfies $z/(z + n) = 4i$ for a positive integer n. Find n.

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2009 AIME I Problem 3: Cancel symmetric binomial coefficients

Eight independent coin flips have head probability p. The chance of three heads is one twenty-fifth of the chance of five heads. Find numerator plus denominator of p in lowest terms.

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2009 AIME I Problem 4: A diagonal meeting a short intercept segment

In parallelogram $ABCD$, points $M \in AB$ and $N \in AD$ satisfy $AM/AB = 17/1000$ and $AN/AD = 17/2009$. If $P = AC \cap MN$, find AC/AP.

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2009 AIME I Problem 6: Integer values of a stepped power function

How many positive integers $N < 1000$ occur as $x^{\lfloor x \rfloor}$ for some real x where the expression is defined?

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2009 AIME I Problem 7: An exponential recurrence that telescopes

Let $a_1 = 1$ and $5^{a_{n+1}-a_n} - 1 = 1/(n + 2/3)$ for $n \geq 1$. Find the least integer $k > 1$ such that a_k is integral.

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2009 AIME I Problem 8: Sum of pairwise differences

Sum all positive differences between pairs from $\{1, 2, 4, \ldots, 2^{10}\}$, and report the remainder modulo 1000.

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2009 AIME I Problem 13: Infinite integrality forces a two-cycle

Every term of a sequence is a positive integer, and $a_{n+2} = (a_n + 2009)/(1 + a_{n+1})$ for $n \geq 1$. Find the smallest possible $a_1 + a_2$.

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2009 AIME I Problem 14: Minimizing a second moment with fixed first and fourth moments

A sequence of 350 entries, each in $\{1, 2, 3, 4\}$, satisfies $\sum a_i = 513$ and $\sum a_i^4 = 4745$. Find the smallest possible $\sum a_i^2$.

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2009 AIME II Problem 1: Equal paint remainders

Initially there are 130 ounces of blue paint, 164 of red, and 188 of white. Four equal-size stripes use blue, red, white, and a red-white mixture, respectively. Equal amounts of the three original colors remain. Find the total paint left.

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2009 AIME II Problem 2: Squared logarithmic exponents

Positive reals satisfy $a^{\log_3 7} = 27$, $b^{\log_7 11} = 49$, and $c^{\log_{11} 25} = \sqrt{11}$. Find $a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11} 25)^2}$.

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2009 AIME II Problem 3: Perpendicular lines in a rectangle

Rectangle $ABCD$ has $AB = 100$. If E is the midpoint of AD and $AC \perp BE$, find the greatest integer less than AD.

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2009 AIME II Problem 4: The smallest winning score

For some number of children, the child ranked k eats exactly $n + 2 - 2k$ grapes, a nonnegative amount. Their total is 2009. Find the smallest possible winning count n.

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2009 AIME II Problem 5: A circle tangent to three interior circles

An equilateral triangle is inscribed in a radius-10 circle. Three internally tangent circles touch the outer circle at the triangle vertices and have radii $3, 2, 2$. A fourth circle is externally tangent to all three. Its radius is reduced m/n. Find $m+n$.

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2009 AIME II Problem 7: A sum with strictly increasing dyadic denominators

Write $\sum_{i=1}^{2009}(2i-1)!!/(2i)!!$ in lowest terms with denominator $2^a b$, where b is odd. Find $ab/10$.

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2009 AIME II Problem 10: Two equal-angle observations

Points A, B, C, D satisfy $AB=5$, $BC=12$, $AC=13$, $\angle BAD = \angle CAD$, and $\angle ACB = \angle DCB$. Write $AD = p\sqrt{q}/r$ in reduced squarefree form and find $p+q+r$.

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2009 AIME II Problem 11: Exactly fifty logarithmic neighbors

Positive integers $m \geq n$ have exactly fifty positive integers k satisfying $|\log m - \log k| < \log n$. Find the sum of all possible products mn.

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2009 AIME II Problem 13: A product of semicircle chords

A radius-2 semicircle with diameter endpoints A, B is divided into seven equal arcs by $C_1, \ldots, C_6$. Let $N = \prod_{k=1}^{6} AC_k \, BC_k$. Find N modulo 1000.

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2009 AIME II Problem 14: A radical recurrence with an eventual two-cycle

Let $a_0 = 0$ and $a_{n+1} = \frac{8}{5}a_n + \frac{6}{5}\sqrt{4^n - a_n^2}$. Find $\lfloor a_{10} \rfloor$.

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2009 AIME II Problem 15: Maximum separation of two chord intercepts

Circle diameter MN has length one. On one semicircle, A is the arc midpoint and $MB = 3/5$. A variable point C lies on the other semicircle. The chords AC, BC meet MN at D, E. If the largest DE is $r - s\sqrt{t}$ in reduced squarefree form, find $r+s+t$.

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2010 AIME I Problem 3: Equal cross powers at a fixed ratio

Positive reals satisfy $y = 3x/4$ and $x^y = y^x$. If $x + y = r/s$ in lowest terms, find $r+s$.

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2010 AIME I Problem 5: Equality in a difference-of-squares bound

Positive integers $a > b > c > d$ satisfy $a + b + c + d = 2010$ and $a^2 - b^2 + c^2 - d^2 = 2010$. How many distinct values can a take?

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2010 AIME I Problem 6: A quadratic trapped between two parabolas

A quadratic polynomial satisfies $x^2 - 2x + 2 \le P(x) \le 2x^2 - 4x + 3$ for every real x, and $P(11) = 181$. Find $P(16)$.

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2010 AIME I Problem 9: A cubic system reduced to one product

Real triples satisfy $x^3 - xyz = 2$, $y^3 - xyz = 6$, $z^3 - xyz = 20$. If the greatest possible $x^3 + y^3 + z^3$ is reduced m/n, find $m + n$.

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2010 AIME I Problem 11: Revolving a triangular inequality region

The region satisfying $|8 - x| + y \le 10$ and $3y - x \ge 15$ is revolved around $3y - x = 15$. Write its volume as $m\pi/(n\sqrt{p})$ in reduced squarefree form and find $m + n + p$.

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2010 AIME I Problem 13: A rectangle and semicircle cut in a fixed area ratio

A rectangle $ABCD$ and a semicircle with diameter AB lie on opposite sides of AB. A line meets the semicircle at N, AB at U, and CD at T, dividing their union in area ratio $1 : 2$. Given $AU = 84$, $UB = 168$, $AN = 126$, write $AD = m\sqrt{n}$ in squarefree form and find $m + n$.

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2010 AIME I Problem 14: A monotone sum of decimal digit counts

For positive integers n, define $f(n) = \sum_{k=1}^{100} \lfloor \log_{10}(kn) \rfloor$. Find the largest n with $f(n) \le 300$.

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2010 AIME I Problem 15: Equal inradii determine a cevian length

Triangle ABC has $AB = 12$, $BC = 13$, $AC = 15$. A point M inside AC makes triangles ABM and BCM have equal inradii. Write $AM/CM = p/q$ in lowest terms and find $p + q$.

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2010 AIME II Problem 5: The norm of three logarithms

Positive numbers satisfy $xyz = 10^{81}$ and $(\log_{10} x)(\log_{10}(yz)) + (\log_{10} y)(\log_{10} z) = 468$. Find $\sqrt{(\log_{10} x)^2 + (\log_{10} y)^2 + (\log_{10} z)^2}$.

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2010 AIME II Problem 6: The first reducible quartic

Find the smallest positive integer n for which $x^4 - nx + 63$ factors into two nonconstant integer-coefficient polynomials.

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2010 AIME II Problem 7: A cubic constrained by conjugate roots

A monic cubic $P(z) = z^3 + az^2 + bz + c$ has real coefficients and roots $w + 3i, w + 9i, 2w - 4$ for some complex w. Find $|a + b + c|$.

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2010 AIME II Problem 9: A regular hexagon bounded by rotated chords

In regular hexagon $ABCDEF$, let G, H, I, J, K, L be the midpoints of AB, BC, CD, DE, EF, FA. The segments AH, BI, CJ, DK, EL, FG bound a smaller regular hexagon. If its area divided by the original area is reduced m/n, find $m + n$.

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2010 AIME II Problem 10: Quadratics modulo interchange of roots

Count quadratic polynomials with integer coefficients and integer roots whose constant coefficient is 2010.

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2010 AIME II Problem 12: Two isosceles triangles with equal area and perimeter

Two noncongruent integer-sided isosceles triangles have equal areas and perimeters, and their base lengths are in ratio $8 : 7$. Find their least possible common perimeter.

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2010 AIME II Problem 13: Two fixed cards share a rank team

Four people draw distinct uniform cards from $1, \ldots, 52$ and the two lowest form one team. Two specified people hold $a, a + 9$. Among choices with their same-team probability at least $1/2$, find numerator plus denominator of the smallest probability.

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2010 AIME II Problem 14: A doubled angle in a right triangle

Right triangle ABC has right angle C, $AB = 4$, and $\angle A < 45^\circ$. A point P on AB satisfies $CP = 1$ and $\angle APC = 2\angle ACP$. If $AP/BP = p + q\sqrt{r}$ in squarefree form, find $p + q + r$.

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2010 AIME II Problem 15: A radical axis evaluated by endpoint powers

Triangle ABC has $AC = 13$, $BC = 14$, $AB = 15$. Let M,N be the midpoints of AC,AB, and let the angle bisectors from B,C meet AC,AB at D,E. The circles AMN and ADE meet again at P. Ray AP meets BC at Q. If $BQ/CQ = m/n$ in lowest terms, find $m - n$.

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2011 AIME I Problem 1: Balancing two acid mixtures

Containers hold 4 liters of 45% acid, 5 liters of 48% acid, and 1 liter of k% acid. Splitting the third container between the first two makes both 50% acid. If the fraction poured into the first is m/n in lowest terms, find $k+m+n$.

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2011 AIME I Problem 2: Parallel segments across a rectangle

In rectangle $ABCD$, $AB=12$ and $BC=10$. Interior points E, F satisfy $BE=9$, $DF=8$, $BE \parallel DF$, and $EF \parallel AB$; the line BE meets side AD. Write $EF = m\sqrt{n} - p$ with squarefree n and positive integers m, n, p, and find $m+n+p$.

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2011 AIME I Problem 3: Coordinates in rotated perpendicular axes

Line L has slope $5/12$ and contains $(24, -1)$. Its perpendicular M contains $(5, 6)$. Use L, M as the new coordinate axes, with positive directions toward these respective points. If $(-14, 27)$ has coordinates (α, β) in this system, find $\alpha+\beta$.

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2011 AIME I Problem 6: An integer value of a parabola

The graph of ax^2+bx+c, with $a>0$, has vertex $(1/4, -9/8)$, and $a+b+c$ is an integer. The smallest possible a is p/q in lowest terms. Find $p+q$.

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2011 AIME I Problem 8: Folding triangular legs of equal height

A triangular sheet has side lengths $23, 27, 30$. Cut off a corner triangle at each vertex by a fold line parallel to the opposite side, with the three corner interiors disjoint. Fold these corners downward through 90° so their vertex tips lie on one plane below the central tabletop. What is the greatest height, written $m\sqrt{n}/p$ in reduced form with squarefree n? Find $m+n+p$.

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2011 AIME I Problem 9: A logarithmic trigonometric equation

For $x \in [0, \pi/2]$, suppose $\log_{24\sin x}(24\cos x) = 3/2$. Find $24\cot^2 x$.

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2011 AIME I Problem 11: Summing a power-residue orbit without its order

Let R be the set of distinct residues of 2^n modulo 1000 for integers $n \geq 0$. Find the remainder modulo 1000 of the sum of the elements of R.

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2011 AIME I Problem 13: The height of a tilted cube

A cube of edge length 10 lies above a plane. Its nearest vertex is A, and the three vertices adjacent to A are at distances $10, 11, 12$ from the plane. Express the distance of A as $(r - \sqrt{s})/t$ in reduced form and find $r + s + t$.

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2011 AIME I Problem 14: Projecting the midpoint square of an octagon

In a regular octagon $A_1 \cdots A_8$, let M_i be the midpoint of $A_i A_{i+1}$ for odd i. Inward rays from M_1, M_3, M_5, M_7 are perpendicular to their neighboring rays. Their successive intersections are B_1, B_3, B_5, B_7, with $B_1 \in R_1 \cap R_3$. If $B_1 B_3$ equals an octagon side, write $\cos(2\angle A_3 M_3 B_1) = m - \sqrt{n}$ and find $m + n$.

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2011 AIME I Problem 15: Bounding integer roots of a depressed cubic

The polynomial $x^3 - 2011x + m$ has three integer roots. Find the sum of their absolute values.

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2011 AIME II Problem 1: Changing the purchased and consumed amounts

A person drinks a reduced fraction m/n of a purchased beverage. If the purchased amount had been halved and the amount drunk doubled, the waste would have been only $2/9$ of the original waste. Find $m + n$.

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2011 AIME II Problem 2: Three equal segments spanning a square

In square $ABCD$, points E and F lie in the interiors of AD and BC, respectively. If $BE = EF = FD = 30$, find the area of the square.

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2011 AIME II Problem 3: Integer angles in arithmetic progression

The interior angle measures of a convex 18-gon, in increasing order, form a strictly increasing arithmetic progression of integers. Find its smallest angle in degrees.

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2011 AIME II Problem 4: A midpoint on an angle bisector

In triangle ABC, $AB = 20$ and $AC = 11$. The internal bisector from A meets BC at D, and M is the midpoint of AD. Line BM meets AC at P. If $CP/PA = m/n$ in lowest terms, find $m + n$.

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2011 AIME II Problem 5: Geometric blocks of equal length

A geometric sequence has sum 200 over its first 2011 terms and sum 380 over its first 4022 terms. Find the sum over its first 6033 terms.

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2011 AIME II Problem 8: Independent choices among rotated complex roots

Let $z_1, \ldots, z_{12}$ be the roots of $z^{12} = 2^{36}$. Independently choose each w_j to equal z_j or iz_j. If the maximum of $\operatorname{Re} \sum_j w_j$ is $m + \sqrt{n}$ for positive integers m, n, find $m + n$.

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2011 AIME II Problem 9: Completing a cyclic cubic to a product

Nonnegative $x_1, \ldots, x_6$ sum to 1 and satisfy $x_1x_3x_5 + x_2x_4x_6 \geq 1/540$. Find $p + q$ if the maximum of $\sum_{j=1}^{6} x_j x_{j+1} x_{j+2}$, with indices taken modulo 6, is the reduced fraction p/q.

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2011 AIME II Problem 10: A diameter through two chord projections

In a circle of radius 25 centered at O, chords of lengths 30 and 14 meet at P. Their midpoints are 12 units apart. If $OP^2 = m/n$ in lowest terms, find the remainder of $m + n$ modulo 1000.

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2011 AIME II Problem 11: A tridiagonal determinant series

For $n \geq 1$, let D_n be the determinant of the n by n matrix with diagonal entries 10, entries 3 immediately above and below the diagonal, and all other entries zero. If $\sum_{n=1}^{\infty} (8D_n + 1)^{-1} = p/q$ in lowest terms, find $p + q$.

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2011 AIME II Problem 13: Circumcenters along a square diagonal

Point P lies inside diagonal AC of square $ABCD$, with $AP > CP$ and side length 12. Let O_1, O_2 be the circumcenters of ABP, CDP. If $\angle O_1 P O_2 = 120^\circ$ and $AP = \sqrt{a} + \sqrt{b}$ for positive integers a, b, find $a + b$.

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2011 AIME II Problem 15: Floor equality on three intervals

Let $P(x) = x^2 - 3x - 9$ and choose x uniformly on $[5, 15]$. The probability that $\lfloor \sqrt{P(x)} \rfloor = \sqrt{P(\lfloor x \rfloor)}$ has form $(\sqrt{a} + \sqrt{b} + \sqrt{c} - d)/e$. Find $a + b + c + d + e$.

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2012 AIME I Problem 2: The sum increase reveals the number of terms

An arithmetic progression has total 715. Increasing its kth term by $2k - 1$ changes the total to 836. Find the sum of the original first, middle, and last terms.

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2012 AIME I Problem 4: A shared horse and the first integer meeting point

Two travelers start together along a road with mileposts at every positive integer distance. Their walking speeds are 4 and 5/2 mph; their shared horse carries either at 6 mph. The slower walker rides first to post one and leaves the horse. Thereafter each rider passes the other traveler, leaves the horse at the next post, and walks on. At their first meeting at a milepost, the distance is n miles and elapsed time is t minutes. Find $n + t$.

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2012 AIME I Problem 6: A positive imaginary part selects a root-of-unity denominator

Complex numbers satisfy $z^{13} = w$ and $w^{11} = z$. Suppose $\operatorname{Im} z = \sin(m\pi/n)$ for coprime positive integers $m < n$. Determine n.

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2012 AIME I Problem 7: Stationary coins on a network

On a connected undirected network, one central vertex has degree five, five vertices have degree three, and ten have degree four. The vertices hold 3360 coins and simultaneously split all coins equally among neighbors; every holding remains unchanged. Find the central holding.

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2012 AIME I Problem 9: Logarithmic coordinates expose a fixed monomial

Positive x, y, z satisfy the defined nonzero common value

$$2\log_x(2y) = 2\log_{2x}(4z) = \log_{2x^4}(8yz) \neq 0.$$

Write $xy^5z = 2^{-p/q}$ with coprime positive integers p, q, and find $p + q$.

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2012 AIME I Problem 13: Equilateral triangles with three prescribed center distances

Three circles have a common center and radii $3, 4, 5$. Among equilateral triangles having one vertex on each circle, maximize the area. Write it as $a + (b/c)\sqrt{d}$, with positive integers, b, c coprime and d squarefree, and find $a + b + c + d$.

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2012 AIME I Problem 14: A zero root sum places the triangle centroid at the origin

The complex roots a, b, c of $z^3 + qz + r$ form a nondegenerate right triangle in the complex plane and satisfy $|a|^2 + |b|^2 + |c|^2 = 250$. If its hypotenuse is h, find h^2. The coefficients q, r need not be real.

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2012 AIME II Problem 2: Eliminate a geometric ratio without finding its sign

Two geometric progressions with the same nonzero common ratio satisfy $a_1 = 27$, $b_1 = 99$, and $a_{15} = b_{11}$. Find a_9. The nonzero-ratio convention is discussed in the source note.

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2012 AIME II Problem 4: Three equal travel times determine a rectangle

From a rectangle's northeast corner, riders of speeds 8.6, 6.2, and 5 m/s start together. The first follows the boundary westward, the second southward, and the third rides straight to a point D on the south side. They first reach D simultaneously. The east-west side is longer. Find the primitive ratio of length, width, and distance from the southeast corner to D, and report its three entries' sum.

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2012 AIME II Problem 6: A complex-distance maximum fixes a fourth power

Choose $z = a + bi$ with $|z| = 5$ and $b > 0$ to maximize $|(1 + 2i)z^3 - z^5|$. If $z^4 = c + di$, find $c + d$.

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2012 AIME II Problem 8: Solve directly for a complex product

Nonzero complex numbers z, w obey $z + 20i/w = 5 + i$ and $w + 12i/z = -4 + 10i$. Find the minimum possible $|zw|^2$.

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2012 AIME II Problem 9: Ratios determine squared trigonometric coordinates

Real angles x, y satisfy $\sin x/\sin y = 3$ and $\cos x/\cos y = 1/2$. Express $\sin(2x)/\sin(2y)+\cos(2x)/\cos(2y)$ as a reduced positive fraction p/q and find $p + q$.

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2012 AIME II Problem 10: A floor-product range is a disjoint union of integer blocks

Count positive integers $n < 1000$ representable as $x\lfloor x \rfloor$ for some positive real x.

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2012 AIME II Problem 11: A fractional-linear map has period three

Let $f(x) = 2/3 - 3/(3x + 1)$, and write $f^{\circ n}$ for its n-fold composition wherever all steps are defined. Solve $f^{\circ 1001}(x) = x - 3$; express the solution as a reduced positive fraction m/n and find $m + n$.

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2012 AIME II Problem 13: Sum squared distances before computing rotated vertices

An equilateral triangle ABC has side $\sqrt{111}$. The two points D_1, D_2 satisfy $AD_i = \sqrt{111}$ and $BD_i = \sqrt{11}$. For each segment AD_i, construct both equilateral third vertices, giving four points $E_1, \ldots, E_4$. Find $\sum_{j=1}^{4} CE_j^2$.

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2013 AIME I Problem 1: Travel time in fixed speed ratios

A race consists of swimming $1/2$ mile, cycling 30 miles, and running 8 miles. The three constant speeds have ratio $1:10:5$. Total time is 255 minutes. Find the cycling time in minutes.

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2013 AIME I Problem 3: Area fractions determine a reciprocal side ratio

Two perpendicular cuts parallel to the sides partition a square into two squares and two nonsquare rectangles. Let a, b be the two segment lengths into which either cut divides its perpendicular side. The two smaller squares occupy $9/10$ of the original area. Find $a/b + b/a$.

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2013 AIME I Problem 5: A cubic hides a difference of cubes

Express the real root of $8x^3 - 3x^2 - 3x - 1 = 0$ as $(\sqrt[3]{a} + \sqrt[3]{b} + 1)/c$ with positive integers a, b, c, and determine $a + b + c$ in this form.

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2013 AIME I Problem 8: A function domain imposes an integer divisor

For positive integers $m > 1, n$, the real domain of $\arcsin(\log_m(nx))$ has length $1/2013$. Find the least possible $m + n$, reduced modulo 1000.

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2013 AIME I Problem 10: Conjugate integer roots and distinct cubics

Consider all cubics $x^3 - ax^2 + bx - 65$ with nonzero integers a, b that have a root $r + si$ for nonzero integers r, s. For each distinct coefficient pair (a, b), take the sum of its three roots. Find the sum of these values over all admissible pairs.

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2013 AIME I Problem 13: A self-similar union sums to one rational area

Begin with a triangle AB_0C_0 of side lengths $AB_0 = 12$, $B_0C_0 = 17$, $AC_0 = 25$. At each stage choose B_n on AB_{n-1} and C_n on AC_{n-1} so that, in the displayed vertex order,

$$AB_nC_n \sim B_{n-1}C_nC_{n-1} \sim AB_{n-1}C_{n-1}.$$

If the union of triangles $B_{n-1}C_nB_n$ for $n \geq 1$ has reduced rational area p/q, determine q.

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2013 AIME I Problem 14: Combine two trigonometric series into one complex series

For $\pi \leq \theta < 2\pi$, define the convergent series

$$P = \frac{\cos\theta}{2} - \frac{\sin 2\theta}{4} - \frac{\cos 3\theta}{8} + \frac{\sin 4\theta}{16} + \cdots,$$

$$Q = 1 - \frac{\sin\theta}{2} - \frac{\cos 2\theta}{4} + \frac{\sin 3\theta}{8} + \frac{\cos 4\theta}{16} + \cdots.$$

The four-term sign and sine/cosine patterns repeat, and the term of frequency k has denominator 2^k. Given $P/Q = 2\sqrt{2}/7$, write $\sin\theta = -m/n$ in lowest terms and find $m+n$.

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2013 AIME II Problem 1: Convert a time by its fraction of a day

A decimal clock divides one day into 10 hours of 100 minutes each. At the instant corresponding to ordinary 6:36 a.m., it reads $A{:}BC$. Determine the three-digit integer $100A + 10B + C$.

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2013 AIME II Problem 2: Nested logarithms impose a power-of-two divisor

For positive integers a, b, suppose $\log_2(\log_{2^a}(\log_{2^b}(2^{1000}))) = 0$. Sum the distinct possible values of $a+b$.

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2013 AIME II Problem 3: Half the burn time is a triangular-number boundary

A candle initially has height 119 cm. Its kth centimeter from the top takes $10k$ seconds to burn. At half of its total burn time, its remaining height is h cm. Determine $10h$.

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2013 AIME II Problem 7: Staff reductions force a unique feasible integer solution

Each clerk sorts 30 files per hour. Starting with x clerks, the same positive number t leaves after each of the first three hours. The remaining group finishes 1775 files at time 3 hours 10 minutes. How many files were sorted during the first 90 minutes?

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2013 AIME II Problem 12: Cubics with roots on two circles

Count monic integer-coefficient cubics all of whose complex roots have modulus either 13 or 20.

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2013 AIME II Problem 15: A triangle identity with a radical-sign branch condition

Triangle angles A, B, C satisfy

$$\cos^2 A + \cos^2 B + 2\sin A \sin B \cos C = 15/8,$$

$$\cos^2 B + \cos^2 C + 2\sin B \sin C \cos A = 14/9.$$

The expression $\cos^2 C + \cos^2 A + 2\sin C \sin A \cos B$ has the form $(p - q\sqrt{r})/s$ for positive integers p, q, r, s, with r squarefree and $\gcd(p+q, s) = 1$. Determine $p+q+r+s$.

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2014 AIME I Problem 4: Two trains passing opposite cyclists

Two cyclists ride east and west at 20 mph. Two equal-length trains travel in opposite directions and each passes the eastbound cyclist in one minute. The westbound train takes ten times as long as the eastbound train to pass the westbound cyclist. If train length is m/n miles in lowest terms, find $m+n$.

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2014 AIME I Problem 6: Two parabolas with a shared axis

The parabolas $y = 3(x-h)^2 + j$ and $y = 2(x-h)^2 + k$ have vertical intercepts 2013 and 2014. Each has two positive integer horizontal intercepts. Find h.

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2014 AIME I Problem 7: The largest argument seen from an exterior point

Complex numbers satisfy $|w| = 1$ and $|z| = 10$. Let $\theta = \arg((w-z)/z)$. If the maximum of $\tan^2\theta$ is p/q in lowest terms, find $p+q$.

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2014 AIME I Problem 9: A cubic designed around a radical

The real roots of $\sqrt{2014}x^3 - 4029x^2 + 2 = 0$ are $x_1 < x_2 < x_3$. Find $x_2(x_1 + x_3)$.

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2014 AIME I Problem 13: A square cut by equal perpendicular chords

Points E, F, G, H lie respectively on sides AB, BC, CD, DA of a square. Segments EG, FH are perpendicular, each of length 34, and meet at P. The areas $AEPH, BFPE, CGPF, DHPG$ have ratio $269 : 275 : 405 : 411$. Find the square area.

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2014 AIME I Problem 14: Centering symmetric rational poles

Let m be the largest real solution of $3/(x-3) + 5/(x-5) + 17/(x-17) + 19/(x-19) = x^2 - 11x - 4$. Write $m = a + \sqrt{b + \sqrt{c}}$ with positive integers a, b, c, and find $a+b+c$.

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2014 AIME I Problem 15: A circle through a right-triangle vertex

In right triangle ABC, $AB = 3, BC = 4, AC = 5$. A circle meets AB at B, E, BC at B, D, and AC at F, G. Suppose $EF = DF$ and $DG/EG = 3/4$. Write $DE = a\sqrt{b}/c$ in reduced radical form and find $a+b+c$.

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2014 AIME II Problem 1: Changing work rates as painters join

Abe paints a room in 15 hours; Bea works 50 percent faster and Coe twice as fast as Abe. Abe works alone for 90 minutes, then Bea joins until half the room is painted, then Coe joins until completion. Find total elapsed minutes.

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2014 AIME II Problem 4: Compatible repeating blocks

Digits a, b, c satisfy $0.\overline{ab} + 0.\overline{abc} = 33/37$. Find the three-digit integer $100a + 10b + c$.

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2014 AIME II Problem 5: Two cubics with translated roots

Two roots of $p(x) = x^3 + ax + b$ are r, s; two roots of $q(x) = x^3 + ax + b + 240$ are $r + 4, s - 3$. Find the sum of all possible values of $|b|$.

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2014 AIME II Problem 7: An alternating logarithmic product

Let $f(x) = (x^2 + 3x + 2)^{\cos(\pi x)}$. Find the sum of positive integers n satisfying $|\sum_{k=1}^{n} \log_{10} f(k)| = 1$.

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2014 AIME II Problem 8: Three mutually tangent circles and a diameter

A radius-2 circle has diameter AB. A smaller circle is internally tangent at A. A third circle is internally tangent to the large circle, externally tangent to the second circle, and tangent to AB. The second radius is three times the third. Write it as $\sqrt{m} - n$ and find $m + n$.

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2014 AIME II Problem 10: A reciprocal equation creates an equilateral triangle

Let $|z| = 2014$. Form the polygon with vertices z and all complex w satisfying $1/(z + w) = 1/z + 1/w$. Its area is $n\sqrt{3}$. Find n modulo 1000.

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2014 AIME II Problem 11: An isosceles extension of a triangle

In triangle RED, angles at R, E are 75 and 45 degrees and $RD = 1$. Let M be the midpoint of RD, and choose C on ED with $RC \perp EM$. Extend DE beyond E to A so that $AC = AR$. If $AE = (a - \sqrt{b})/c$ in reduced form, find $a + b + c$.

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2014 AIME II Problem 12: A triple-angle condition forces one angle

Triangle angles satisfy $\cos 3A + \cos 3B + \cos 3C = 1$. Two sides have lengths 10 and 13. The greatest possible third side is $\sqrt{m}$; find m.

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2015 AIME I Problem 1: Pairing two alternating product sums

Let $A = 1 \cdot 2 + 3 \cdot 4 + \cdots + 37 \cdot 38 + 39$ and $B = 1 + 2 \cdot 3 + 4 \cdot 5 + \cdots + 38 \cdot 39$. Find $|A - B|$.

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2015 AIME I Problem 9: When a multiplicative recurrence reaches zero

For $1 \le x, y, z \le 10$ initialize $a_1 = x, a_2 = y, a_3 = z$ and $a_n = a_{n-1}|a_{n-2} - a_{n-3}|$. Count initial triples whose sequence eventually reaches zero.

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2015 AIME I Problem 10: A cubic attaining only two levels

A real cubic polynomial f satisfies $|f(j)| = 12$ for $j = 1, 2, 3, 5, 6, 7$. Find $|f(0)|$.

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2015 AIME I Problem 13: A product over all odd degree angles

Write $\prod_{k=1}^{45} \csc^2((2k-1)^\circ) = m^n$ with integers $m, n > 1$. Find $m + n$.

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2015 AIME I Problem 14: Parity of piecewise-linear area

Let $A(n)$ be the area under $y = x\lfloor\sqrt{x}\rfloor$ from $x = 1$ to $x = n$. Count $2 \le n \le 1000$ for which $A(n)$ is integral.

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2015 AIME II Problem 4: Logarithmic lengths form a familiar triangle

An isosceles trapezoid has bases $\log 3, \log 192$ and altitude $\log 16$, with a common logarithm base greater than 1. Its perimeter is $\log(2^p 3^q)$. Find $p + q$.

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2015 AIME II Problem 6: Two possible cubics with positive integer roots

The polynomial $2x^3 - 2ax^2 + (a^2 - 81)x - c$ has three positive integer roots, counted with multiplicity. Knowing a still leaves exactly two possible values of c. Find their sum.

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2015 AIME II Problem 8: An integer inequality bounds a cubic quotient

Positive integers a, b satisfy $(ab + 1)/(a + b) < 3/2$. If the largest possible value of $(a^3b^3 + 1)/(a^3 + b^3)$ is p/q in lowest terms, find $p + q$.

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2015 AIME II Problem 13: Negative partial sums of successive sines

For $n \ge 1$, let $a_n = \sum_{k=1}^{n} \sin k$, with angles in radians. Find the index of the hundredth negative term.

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2015 AIME II Problem 14: Extract the requested symmetric quantity directly

Real numbers x, y satisfy $x^4y^5 + y^4x^5 = 810$ and $x^3y^6 + y^3x^6 = 945$. Find $2x^3 + (xy)^3 + 2y^3$.

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2015 AIME II Problem 15: Equal areas cut from two tangent circles

Two externally tangent circles have radii 1 and 4 and meet at A. A common external tangent touches them at B, C. A line through A meets the circles again at D, E, respectively. If B, C lie on the same side of the line and $[DBA] = [ACE] = m/n$ in lowest terms, find $m + n$.

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2016 AIME I Problem 1: Opposite-ratio geometric series

For $|r| < 1$, let $S(r) = 12 + 12r + 12r^2 + \cdots$. If $S(a)S(-a) = 2016$, find $S(a) + S(-a)$.

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2016 AIME I Problem 5: Two arithmetic totals share a day count

A reader starts with n pages in t minutes, then increases both quantities by one each day. She reads on at least two days, finishing 374 pages in 319 minutes; n, t are positive integers. Find $n + t$.

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2016 AIME I Problem 7: Real values of a complex radical expression

Using principal square roots, count ordered integer pairs (a, b) for which $\sqrt{ab + 2016}/(ab + 100) - i\sqrt{|a + b|}/(ab + 100)$ is defined and real.

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2016 AIME I Problem 9: A rectangle optimization with a segment inconsistency

Triangle ABC has $AB = 40$, $AC = 31$, and $\sin A = 1/5$. It lies in rectangle $AQRS$ with B on segment QR and C on segment RS. Determine the maximum rectangle area. The reported contest answer is 744; the segment conditions require the correction below.

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2016 AIME I Problem 10: Alternating geometric and arithmetic triples

A strictly increasing positive integer sequence has every triple $a_{2k-1}, a_{2k}, a_{2k+1}$ geometric and every triple $a_{2k}, a_{2k+1}, a_{2k+2}$ arithmetic. Given $a_{13} = 2016$, find a_1.

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2016 AIME I Problem 11: A polynomial shift equation

A nonzero polynomial satisfies $(x - 1)P(x + 1) = (x + 2)P(x)$ for every real x and $P(2)^2 = P(3)$. If $P(7/2) = m/n$ in lowest terms, find $m + n$.

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2016 AIME II Problem 1: A geometric allocation becomes arithmetic

Alex, Betty, and Charlie initially have 444 peanuts in a strictly increasing geometric progression. They eat 5,9,25 peanuts respectively, after which their counts, in the same order, form an arithmetic progression. Find Alex's initial count.

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2016 AIME II Problem 3: Three logarithms share one product

Real x, y, z satisfy $\log_2(xyz - 3 + \log_5 x) = 5$, $\log_3(xyz - 3 + \log_5 y) = 4$, and $\log_4(xyz - 3 + \log_5 z) = 4$. Find $|\log_5 x| + |\log_5 y| + |\log_5 z|$.

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2016 AIME II Problem 5: An infinite chain of right-triangle altitudes

Right triangle ABC_0 has pairwise coprime positive integer sides and perimeter p. Let C_1 be the foot from C_0 to AB; for $n \geq 2$, let C_n be the foot from C_{n-1} to $C_{n-2}B$. If $C_0C_1 + C_1C_2 + \cdots = 6p$, find p.

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2016 AIME II Problem 6: Absolute coefficient sum by parity

For $P(x) = 1 - x/3 + x^2/6$ and $Q(x) = P(x)P(x^3)P(x^5)P(x^7)P(x^9)$, find numerator plus denominator of the sum of absolute coefficients of Q.

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2016 AIME II Problem 9: An arithmetic sequence plus a geometric sequence

Increasing positive integer sequences satisfy $a_1 = b_1 = 1$, with a_n arithmetic and b_n geometric. Put $c_n = a_n + b_n$. If $c_{k-1} = 100$ and $c_{k+1} = 1000$, find c_k.

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2016 AIME II Problem 15: An apparent maximum forced by Cauchy equality

Set $a_i = 2^{-i}$ for $1 \leq i \leq 215$ and $a_{216} = 2^{-215}$. Positive numbers x_i satisfy $\sum x_i = 1$ and $\sum_{i<j} x_i x_j = 107/215 + \sum_i a_i x_i^2/[2(1 - a_i)]$. If the greatest possible $x_2 = m/n$ in lowest terms, find $m + n$.

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2017 AIME I Problem 10: A real cross ratio traces a circle

Let $z_1 = 18 + 83i$, $z_2 = 18 + 39i$, and $z_3 = 78 + 99i$. Among complex $z \neq z_3$ making $\frac{z_3 - z_1}{z_2 - z_1} \frac{z - z_2}{z - z_3}$ real, take the one with greatest imaginary part. Find its real part.

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2017 AIME I Problem 14: Nested logarithms become a monotone power equation

For $a > 1, x > 1$, suppose $\log_a(\log_a(\log_a 2) + \log_a 24 - 128) = 128$ and $\log_a(\log_a x) = 256$. Find x modulo 1000.

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2017 AIME I Problem 15: The smallest equilateral triangle between three sides

A right triangle has legs 5 and $2\sqrt{3}$. An equilateral triangle has one vertex on each side. Its least possible area is $m\sqrt{p}/n$, where m/n is reduced and p squarefree. Find $m+n+p$.

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2017 AIME II Problem 5: Four pair sums force a complementary pair

Four distinct real numbers have six pairwise sums. Four of these are 189,320,287,234; the remaining two are x, y. Find the greatest possible $x+y$.

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2017 AIME II Problem 7: Count parameters after enforcing the logarithm domain

Count integers $k \in [-500, 500]$ for which $\log(kx) = 2\log(x+2)$ has exactly one real solution.

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2017 AIME II Problem 12: The limit point of rotating nested circles

Starting with a unit circle centered at O, choose A_0 on it. For $n \geq 1$, a circle of radius r^n is internally tangent to the preceding circle at A_{n-1}, and A_n is 90 degrees counterclockwise from that tangency point on the new circle. For $r = 11/60$, write the distance from O to the common limit point as m/n and find $m+n$.

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2017 AIME II Problem 15: The geometric median of a disphenoid

A tetrahedron has opposite edge pairs $AD = BC = 28$, $AC = BD = 44$, and $AB = CD = 52$. If the least possible value of $AX + BX + CX + DX$ over points X in space is $m\sqrt{n}$ with squarefree n, find $m+n$.

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2018 AIME I Problem 1: Integer-root quadratics counted by a sum

Count modulo 1000 pairs (a, b) with $1 \leq a \leq 100, b \geq 0$ for which $x^2 + ax + b$ splits over the integers.

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2018 AIME I Problem 5: Two logarithmic equations restrict a homogeneous ratio

For real (x, y) satisfying $\log_2(2x+y) = \log_4(x^2+xy+7y^2)$, choose K so that $\log_3(3x+y) = \log_9(3x^2+4xy+Ky^2)$. Find the product of all possible K.

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2018 AIME I Problem 6: Two disjoint root-of-unity families

Count complex numbers z satisfying $|z| = 1$ and $z^{720} - z^{120} \in \mathbb{R}$. Find the count modulo 1000.

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2018 AIME II Problem 1: A return trip has twice the meeting time

Points A, B, C occur in order on a path, with $AC = 1800$ meters. Eve runs at speed v, Ina at $2v$, Paul at $4v$. Ina starts at A toward C, Paul at B toward C, and Eve at C toward A. Paul turns upon meeting Eve; Paul and Ina reach B together. Find AB.

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2018 AIME II Problem 5: Recover a norm without choosing square-root branches

Complex numbers satisfy $xy = -80 - 320i$, $yz = 60$, and $zx = -96 + 24i$. If $x + y + z = a + bi$ with real a,b, find $a^2 + b^2$.

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2018 AIME II Problem 6: Random parameter with four real roots

For a chosen uniformly in $[-20, 18]$, find numerator plus denominator of the probability that $x^4 + 2ax^3 + (2a - 2)x^2 + (-4a + 3)x - 2$ has only real roots.

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2018 AIME II Problem 12: An area balance bisects one diagonal

A convex quadrilateral has $AB = CD = 10$, $BC = 14$, $AD = 2\sqrt{65}$. Its diagonals meet at P, with $[APB] + [CPD] = [BPC] + [APD]$. Find its area.

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2019 AIME I Problem 4: Ordered substitutions

A team has 11 active players and 11 unused substitutes. Count modulo 1000 histories with at most 3 successive substitutions, with removed players unable to return.

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2019 AIME I Problem 7: Prime exponents hidden in gcd and lcm logarithms

Positive integers x,y satisfy $\log_{10} x + 2\log_{10} \gcd(x, y) = 60$ and $\log_{10} y + 2\log_{10} \operatorname{lcm}(x, y) = 570$. If m,n count the prime factors of x,y with multiplicity, find $3m + 2n$.

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2019 AIME I Problem 8: Two-variable power sums from one product

For real x, $\sin^{10} x + \cos^{10} x = 11/36$. If $\sin^{12} x + \cos^{12} x = m/n$ in lowest terms, find $m + n$.

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2019 AIME I Problem 10: Read coefficients before expanding a large cube

Distinct complex numbers $z_1, \ldots, z_{673}$ satisfy $\prod_j (x - z_j)^3 = x^{2019} + 20x^{2018} + 19x^{2017} + g(x)$, where $\deg g \le 2016$. Write $|\sum_{j<k} z_j z_k| = m/n$ in lowest terms and find $m + n$.

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2019 AIME I Problem 12: A right angle becomes a purely imaginary quotient

Let $f(z) = z^2 - 19z$. Complex numbers z,f(z),f(f(z)) form a nondegenerate right triangle, with right angle at f(z). One admissible z is $m + \sqrt{n} + 11i$ for positive integers m,n, with n squarefree. Find $m+n$.

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2019 AIME II Problem 3: Overlapping products of seven integers

Count positive integer tuples (a, b, c, d, e, f, g) satisfying $abc = 70, cde = 71, efg = 72$.

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2019 AIME II Problem 6: A nested logarithm is a second linear variable

All unmarked logarithms have an unknown base $b \geq 2$. For $x > 1$, $3\log_b(\sqrt{x}\log_b x) = 56$ and $\log_{\log_b x} x = 54$. Find b.

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2019 AIME II Problem 8: A coefficient bound attained at a sixth root of unity

Real numbers $a, b, c \leq 2019$ define $f(z) = az^{2018} + bz^{2017} + cz^{2016}$. If $f((1+\sqrt{3}i)/2) = 2015 + 2019\sqrt{3}i$, find the remainder of $f(1)$ upon division by 1000.

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2019 AIME II Problem 10: A tangent sign itinerary fixes every binary digit

An angle $0 < \theta < 90^\circ$ has $\tan(2^n\theta) > 0$ precisely when $3 \mid n$, and negative otherwise, for every integer $n \geq 0$. Write $\theta = p/q$ degrees in lowest terms and find $p + q$.

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2019 AIME II Problem 15: Two chord powers recover both sides of a triangle

In acute triangle ABC, P is the perpendicular foot from C to AB and Q the foot from B to AC. Line PQ meets the circumcircle at X,Y in the order X,P,Q,Y, with $XP = 10$, $PQ = 25$, $QY = 15$. If $AB \cdot AC = m\sqrt{n}$ with n squarefree, find $m + n$.

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2020 AIME I Problem 2: A geometric progression with negative logarithm terms

The numbers $\log_8(2x)$, $\log_4 x$, $\log_2 x$ form a geometric progression with positive ratio. For the unique positive x, write $x = m/n$ reduced and find $m + n$.

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2020 AIME I Problem 6: Equal spheres resting in unequal circular holes

Two circular holes of radii 1 and 2 have centers 7 units apart in a horizontal board. Two equal spheres rest in the holes, with their centers above the board, and touch each other. If their common radius squared is m/n reduced, find $m + n$.

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2020 AIME I Problem 8: A shrinking spiral is a complex geometric series

A bug starts at O and walks 5 units east. Before each later walk it turns 60 degrees counterclockwise and halves its walking distance. Its positions converge to P. If $OP^2 = m/n$ reduced, find $m+n$.

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2020 AIME I Problem 11: Counting coefficients through common images

For integer a, b, c in $[-10, 10]$, set $f(x) = x^2 + ax + b$ and $g(x) = x^2 + cx + d$. Count triples for which an integer d makes $g(f(2)) = g(f(4)) = 0$.

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2020 AIME I Problem 14: Two fibers of a monic quadratic

A monic quadratic P with complex coefficients has four distinct solutions 3,4,a,b to $P(P(x)) = 0$. Find the sum of all possible values of $(a+b)^2$.

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2020 AIME I Problem 15: An orthocenter tangent determines a chord and an altitude

An acute triangle ABC has orthocenter H and circumcircle omega. The tangent at H to the circumcircle of HBC meets omega at X,Y. Given $HA = 3$, $HX = 2$, $HY = 6$, write $[ABC] = m\sqrt{n}$ with n squarefree and find $m+n$.

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2020 AIME II Problem 3: Cancel a common logarithm factor

If $\log_{2^x}(3^{20}) = \log_{2^{x+3}}(3^{2020})$, write x=m/n reduced and find m+n.

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2020 AIME II Problem 6: A rational recurrence with a five-step return

Set $t_1 = 20, t_2 = 21$ and $t_n = (5t_{n-1} + 1)/(25t_{n-2})$ for n≥3. Write $t_{2020} = p/q$ reduced and find p+q.

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2020 AIME II Problem 8: Zeros of nested absolute values form a parity interval

Let $f_1(x) = |x-1|$ and $f_n(x) = f_{n-1}(|x-n|)$ for n>1. Find the least n for which the sum of the distinct real zeros of f_n exceeds 500000.

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2020 AIME II Problem 11: Three pairwise sums of monic quadratics

Let $P(x) = x^2 - 3x - 7$. Monic quadratics Q,R have the property that each pair among P+Q,P+R,Q+R has a common root, with three distinct common roots. Given Q(0)=2, write R(0)=m/n reduced and find m+n.

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2020 AIME II Problem 13: Tangent lengths and a monotone angle sum

A convex pentagon ABCDE has an incircle and side lengths $AB = 5$, $BC = CD = DE = 6$, $EA = 7$. Find its area.

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2020 AIME II Problem 14: Iterated fractional-part preimages

Define $f(x) = x\{x\}$, where $\{x\} = x - \lfloor x \rfloor$. Count modulo 1000 the solutions of $f(f(f(x))) = 17$ in $[0, 2020]$.

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2021 AIME I Problem 5: Integer arithmetic progressions satisfying a square identity

A strictly increasing three-term integer arithmetic progression is special when the sum of its squared terms equals its middle term times the square of its common difference. Find the sum of the final terms of all special progressions.

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2021 AIME I Problem 7: Simultaneous sine maxima

Count $1 \le m < n \le 30$ for which some real x satisfies $\sin(mx) + \sin(nx) = 2$.

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2021 AIME I Problem 8: Counting levels with twelve preimages

Count integers c for which $||20|x| - x^2| - c| = 21$ has 12 distinct real solutions.

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2021 AIME I Problem 15: Perpendicular parabolas share a circle of intersection

For positive integer k, consider $y = x^2 - k$ and $x = 2(y-20)^2 - k$. Let S contain k for which the curves meet at four distinct points on a circle of radius at most 21. Find the sum of the least and greatest members of S.

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2021 AIME II Problem 4: Missing coefficients determine a shared conjugate pair

Real-coefficient cubics $x^3 + ax + b$ and $x^3 + cx^2 + d$ have roots-20 and-21 respectively, and share a root $m + i\sqrt{n}$ with positive integers m,n. Find m+n.

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2021 AIME II Problem 7: Partial symmetric sums determine variables in reverse order

Real a,b,c,d satisfy $a + b = -3$, $ab + bc + ca = -4$, $abc + bcd + cda + dab = 14$, and $abcd = 30$. Write $a^2 + b^2 + c^2 + d^2 = m/n$ reduced and find m+n.

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2021 AIME II Problem 15: Walking to the next square with two step sizes

For positive integers n, define f(n)=$\sqrt{n}$ at squares and f(n)=1+f(n+1) otherwise. Define g(n)=$\sqrt{n}$ at squares and g(n)=2+g(n+2) otherwise. Find the least n for which f(n)/g(n)=4/7.

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2022 AIME I Problem 1: Cancel the quadratic terms before evaluating

Quadratics P,Q have leading coefficients 2 and−2, respectively, and both graphs pass through (16,54) and (20,53). Find P(0)+Q(0).

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2022 AIME I Problem 4: Roots-of-unity exponent pairs

Let $w = (\sqrt{3} + i)/2$ and $z = (-1 + i\sqrt{3})/2$. Count $1 \le r, s \le 100$ with $iw^r = z^s$.

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2022 AIME I Problem 5: Subtract squared swimming velocities

A 264-foot-wide river flows east at 14 feet per minute. Melanie, swimming at 80 feet per minute relative to the water, starts D feet downstream from Sherry, whose swimming speed is 60. They swim straight to the same opposite-bank point, equally far from their starting points, and arrive together. Find D.

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2022 AIME I Problem 15: Eliminate radicals while retaining their signs

Positive x,y,z satisfy $\sqrt{2x-xy}+\sqrt{2y-xy} = 1$, $\sqrt{2y-yz}+\sqrt{2z-yz} = \sqrt{2}$, and $\sqrt{2z-zx}+\sqrt{2x-zx} = \sqrt{3}$. Write $[(1-x)(1-y)(1-z)]^2 = m/n$ reduced; find m+n.

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2022 AIME II Problem 4: Divide exponentiated logarithm identities

For positive x other than 1/20 and 1/2, $\log_{20x}(22x) = \log_{2x}(202x)$. Their common value is $\log_{10}(m/n)$ with m,n coprime positive integers. Find m+n.

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2022 AIME II Problem 6: Sharp bounds for two ordered coordinates

Reals $x_1 \le \cdots \le x_{100}$ satisfy $\sum |x_i| = 1$ and $\sum x_i = 0$. Write the maximum of $x_{76} - x_{16}$ as m/n in lowest terms; find m+n.

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2022 AIME II Problem 11: Two angle-bisector reflections determine a quadrilateral

A convex quadrilateral ABCD has AB=2,AD=7,CD=3. The acute angles DAB and ADC have internal bisectors meeting at the midpoint M of BC. Find the square of its area.

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2022 AIME II Problem 12: Pair the foci of two intersecting ellipses

Reals a>4,b>1,x,y satisfy $x^2/a^2 + y^2/(a^2-16) = 1$ and $(x-20)^2/(b^2-1) + (y-11)^2/b^2 = 1$. Find the minimum of a+b.

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2022 AIME II Problem 13: A coefficient reduced to stars and bars

Find coefficient x^{2022} in $(x^{2310}-1)^6/[(x^{105}-1)(x^{70}-1)(x^{42}-1)(x^{30}-1)]$.

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2023 AIME I Problem 2: Two logarithms reduce to one variable

Positive b,n with b≠1 satisfy $\sqrt{\log_b n} = \log_b \sqrt{n}$ and $b \log_b n = \log_b(bn)$. Write n=j/k reduced and find j+k.

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2023 AIME I Problem 9: A cubic with one other equal-value integer

Count monic cubics $x^3 + ax^2 + bx + c$ with a, b, c in $[-20, 20]$ for which exactly one integer m unequal 2 satisfies $p(m) = p(2)$.

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2023 AIME I Problem 10: Bound the floor error before summing residues

For a positive integer a, put $U = \sum_{n=1}^{2023} \lfloor (n^2 - na)/5 \rfloor$. Exactly one a makes −1000<U<1000. Find a+U.

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2023 AIME I Problem 13: A Gram determinant distinguishes two rhombohedra

Every face of each of two noncongruent parallelepipeds is a rhombus with diagonals $\sqrt{21}$ and $\sqrt{31}$. Write the larger volume divided by the smaller as m/n reduced; find m+n.

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2023 AIME I Problem 15: A bounded Gaussian-integer search with all sign cases

Find the largest prime p<1000 for which a Gaussian integer z has absolute value $\sqrt{p}$ and p,Re(z cubed),Im(z cubed) are positive side lengths of a triangle.

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2023 AIME II Problem 1: Pair opposite terms of an arithmetic progression

Six tree yields form an arithmetic progression, with largest twice smallest and total 990. Find the largest yield.

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2023 AIME II Problem 4: Subtract a symmetric quadratic system

Real x,y,z satisfy xy+4 z=yz+4 x=zx+4 y=60. Find the sum of squares of all distinct possible x values.

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2023 AIME II Problem 8: Conjugate pairs collapse a seventh-root product

Let omega be a primitive seventh root of unity. Evaluate $\prod_{k=0}^{6}(\omega^{3k}+\omega^{k}+1)$.

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2023 AIME II Problem 13: Prove the integrality class before counting last digits

For acute A, tan A=2 cos A. Count positive n≤1000 for which sec to the nth power A plus tan to the nth power A is an integer ending in 9.

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2024 AIME I Problem 1: Subtract travel times to eliminate a fixed stop

A 9-kilometer walk plus a fixed coffee stop takes 4 hours at speed s km/h and 2 hours 24 minutes at speed s+2. How many minutes does the same trip take at speed s+1/2?

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2024 AIME I Problem 2: Multiply reciprocal logarithms

Reals x,y>1 satisfy $\log_x(y^x)=\log_y(x^{4y})=10$. Find xy.

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2024 AIME I Problem 7: A reciprocal becomes a conjugate on a circle

For complex z with |z|=4, find the largest possible real part of $(75+117i)z+(96+144i)/z$.

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2024 AIME I Problem 9: A rhombus approaches a hyperbola's asymptote

Four points on $x^2/20-y^2/24=1$ form a rhombus with diagonal intersection at the origin. Find the greatest real number strictly below BD squared for every such rhombus.

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2024 AIME I Problem 12: Count branch pairs with a rigorous curvature certificate

Define $f(t)=||t|-1/2|$ and $g(t)=||t|-1/4|$. How many real pairs satisfy $y=4g(f(\sin(2\pi x)))$ and $x=4g(f(\cos(3\pi y)))$?

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2024 AIME I Problem 15: A double-root boundary gives the largest box diagonal

Consider all rectangular boxes of surface area 54 and volume 23. Let r be the least radius such that each box fits inside a sphere of radius r. Write r squared=p/q reduced; find p+q.

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2024 AIME II Problem 3: Digits constrained by row and column sums

Fill a 2×3 array with decimal digits, allowing leading zeroes. The two three-digit rows sum to 999, and the three two-digit columns sum to 99. Count fillings.

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2024 AIME II Problem 4: Pair logarithm equations to isolate variables

Positive x,y,z satisfy $\log_2(x/(yz)) = 1/2$, $\log_2(y/(xz)) = 1/3$, $\log_2(z/(xy)) = 1/4$. Write $|\log_2(x^4y^3z^2)| = m/n$ reduced; find m+n.

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2024 AIME II Problem 11: A symmetric cubic counting problem

Count ordered nonnegative integer triples with $a+b+c = 300$ and $a^2b + a^2c + b^2a + b^2c + c^2a + c^2b = 6000000$.

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2024 AIME II Problem 12: The unique ladder through a point touches an astroid

Let A=(1/2,0),B=$(0,\sqrt{3}/2)$. Among unit segments joining the positive coordinate axes, there is a unique interior point C of AB lying on no other such segment. Write OC squared=p/q reduced, where O is the origin; find p+q.

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2024 AIME II Problem 13: Evaluate a root product at two conjugate constants

Let omega≠1 satisfy omega to the 13 th power=1. Find the product of $2 - 2\omega^k + \omega^{2k}$ for k=0 through 12, modulo 1000.

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2025 AIME I Problem 8: A circle has exactly one point on each tangent

For real k, the equations $|25 + 20i - z| = 5$ and $|z - 4 - k| = |z - 3i - k|$ have exactly one complex solution z. Write the sum of all possible k as m/n reduced; find m+n.

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2025 AIME I Problem 9: Use reflection symmetry to construct the required intersection

Rotate the parabola y=x squared−4 by 60 degrees counterclockwise about the origin. Its unique fourth-quadrant intersection with the original has y-coordinate $(a - \sqrt{b})/c$, with a,c coprime positive integers. Find a+b+c.

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2025 AIME I Problem 11: Pair sawtooth intersections by Vieta

Define f periodically with period 4 by f(x)=x on $-1 \le x < 1$ and f(x)=2−x on $1 \le x < 3$. Intersect its graph with x=34 y squared. Write the sum of all intersection heights as $(a+b\sqrt{c})/d$ in normalized squarefree form; find a+b+c+d.

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2025 AIME I Problem 12: Order chambers divide an equilateral simplex

In the plane x+y+z=75, the strict inequalities $x-yz < y-zx < z-xy$ describe three disjoint convex regions, exactly one of finite area. Write that finite area as a $\sqrt{b}$ with b squarefree; find a+b.

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2025 AIME II Problem 4: Telescope both the logarithms and their exponents

Write $\prod_{k=4}^{63} \frac{\log_k(5^{k^2-1})}{\log_{k+1}(5^{k^2-4})} = m/n$ in lowest terms. Find m+n.

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2025 AIME II Problem 9: Count sine levels and identify double zeros

Let n count roots of $\sin(7\pi \sin(5x)) = 0$ in $0 < x < 2\pi$, and t count those roots where the graph is tangent to the x-axis. Find n+t.

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2025 AIME II Problem 12: Equal-area fan triangles alternate their radial lengths

A nonconvex 11-gon A 1,...,A 11 has perimeter 20. For i=2 through 10, triangle Ai,A 1,A(i+1) has area 1 and cosine of its angle at A 1 equal 12/13. Write A 1 A 2+A 1 A 11 as $(m\sqrt{n}-p)/q$ in normalized squarefree form; find m+n+p+q.

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2025 AIME II Problem 13: Track reduction before simplifying a rational recurrence

Define $x_1 = 25/11$ and $x_{k+1} = (x_k + 1/x_k - 1)/3$. If x 2025=m/n reduced with positive integers m,n, find (m+n) modulo 1000.

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2025 AIME II Problem 14: An equilateral triangle inside a right triangle controls area

Right triangle ABC has $\angle A = 90^\circ$ and $BC = 38$. Interior points K, L satisfy $AK = AL = BK = CL = KL = 14$. The area of $BKLC$ is $n\sqrt{3}$; find n.

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2025 AIME II Problem 15: Two equal minima force a squared quadratic

For positive k define $f(x) = (x-18)(x-72)(x-98)(x-k)/x$ on x>0. Exactly three k give a minimum attained at two distinct positive x values. Find their sum.

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2026 AIME I Problem 1: Three staggered starts and a common arrival

Patrick walks from school to a park. Tanya starts one hour later, running 2 mph faster; Jose starts another hour later, cycling 7 mph faster than Tanya. All travel at constant speeds and arrive together. If the distance is m/n miles in lowest terms, find $m+n$.

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2026 AIME I Problem 6: A logarithmic quadratic determines a product

Let P be the product of all positive real solutions of $\sqrt[20]{x^{\log_{2026} x}} = 26x$. Find the number of positive divisors of P.

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2026 AIME I Problem 12: Sphere radii form an affine function

In a plane, $AB = 6$, $AC = 4$, and $\angle BAC = 90^\circ$. Reflect the centroid of ABC across BC to obtain D. Spheres of radii $1, 2, 3, r$ touch the plane at A, B, C, D on the same side. All also touch a second plane on the same side of it. Find numerator plus denominator of r in lowest terms.

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2026 AIME I Problem 13: Binomial residue sums over a finite field

For $0 \le r < 502$, let S_r sum $\binom{10000}{k}$ over $0 \le k \le 10000$ congruent to r modulo 502. How many S_r are divisible by 503?

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2026 AIME I Problem 14: Closure controls an equiangular pentagon

An equiangular pentagon has side-length squares summing to 308 and diagonal-length squares summing to 800. Its perimeter squared is $m\sqrt{n}$, where m, n are positive integers and n is squarefree. Find $m+n$.

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2026 AIME II Problem 1: Arithmetic progressions indexed by common divisors

Find the sum of the eleventh terms of all integer arithmetic sequences beginning with 4 and containing both 24 and 34. An alternate archived version asks for the tenth terms instead.

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2026 AIME II Problem 6: Normal distances from a point to a parabola

Find the sum of all radii of circles centered at $(4, 39)$ that are tangent to the parabola $2y = x^2 - 8x + 12$ at at least one point.

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2026 AIME II Problem 8: An integer isosceles triangle with a tiny incenter triangle

In an isosceles triangle ABC, $AB = BC$ and I is the incenter. The perimeters of ABC and AIC have ratio $125 : 6$. All side lengths of both triangles are integers. Find the least possible AB.

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2026 AIME II Problem 9: A divisor-generating decimal with controlled carries

Let $S = \sum_{k\geq 1} 1/(10^k - 1)$. Find $\lfloor 10^{100} S \rfloor$ modulo 1000.

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2026 AIME II Problem 11: Seven signed square-root sums force a zero sum

Find the greatest integer n such that $x^3 - (n/6)x^2 + (n-11)x - 400$ has roots $\alpha^2, \beta^2, \gamma^2$ and the choices of square roots yield exactly seven distinct values of $\alpha + \beta + \gamma$.

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